Deep Learning–Based Neural Decoder for EEG Motor Imagery

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Code, data and per-fold results: github.com/sathvikparangi907-web/neural-decoder-eeg-motor-imagery

***Abstract*—Motor-imagery brain–computer interfaces (BCIs) still require each new user to record a labelled calibration session, and recent decoders add attention and hundreds of thousands of parameters without a common test of whether this helps people a model has never seen. We report a leave-one-subject-out (LOSO) study on BCI Competition IV-2a (nine subjects, four classes) in which six decoders with 3,188 to 697,412 parameters share one preprocessing pipeline and one fold structure, and design choices are judged on a held-out validation subject. Under our first protocol the four strongest deep decoders lay within 2.2 points of each other (50.2–52.4%). Separating early stopping from model selection then closed a 6.9-point gap between validation and test accuracy. In a component study of a convolution–Transformer hybrid, HCT-Net, Euclidean alignment helped and attention did not; recomputing batch-normalisation statistics from the new user's unlabelled trials improved all nine test subjects (+4.3 points). Shrinking HCT-Net from 20,996 to 3,924 parameters cost nothing: it reached 53.6% (κ = 0.381) on unseen users, level with EEGNet (54.3%; difference −0.7 points, 95% CI −4.3 to 2.9). Riemannian baselines with re-centring reached 47.1–50.5%. Familiar users were decoded about 18 points better at every model size. With nine subjects, differences of a few points cannot be resolved, and the largest baselines were evaluated only under the first protocol; within these limits, we found no evidence that extra capacity helps cross-subject decoding on this dataset. Code, data checksums and per-fold logs are public.**

***Index Terms*—Brain–computer interface, cross-subject decoding, electroencephalography, Euclidean alignment, motor imagery, test-time adaptation.**

I. Introduction

Brain–computer interfaces (BCIs) let a person control a computer or another machine directly through recorded brain signals, bypassing the muscles that would normally carry out the intention [1]. Motor imagery (MI) is one of the most widely studied non-invasive paradigms because it does not depend on external stimulation: the user mentally rehearses a movement of the left or right hand, the feet or the tongue, and the decoder has to recognise the resulting change in the sensorimotor rhythms measured by scalp electroencephalography (EEG) [2]. The underlying physiology is well documented. Rehearsing a movement weakens the 8–13 Hz mu rhythm and the 13–30 Hz beta rhythm above the part of the cortex that would control that movement, a phenomenon called event-related desynchronisation (ERD) [3]. The most direct beneficiaries are people with severe motor disabilities who cannot use a keyboard, a touchscreen or their voice; neurorehabilitation and hands-free control are further uses.

In practice the main barrier is calibration. A decoder that performs well for the person whose data it was trained on tends to perform badly for everyone else, so each new user has to provide a labelled calibration recording before the system can be used [4]. The root of the problem is inter-individual variability: anatomy such as skull thickness and cortical folding, the exact electrode positions and the imagery strategy chosen by the user all vary, and the statistics of the recorded signal vary with them [5], [6]. For a substantial share of users an MI-BCI never becomes reliable, which is often referred to as BCI inefficiency [7]. Recent results show how large the effect is. CTNet, a convolution–Transformer decoder, reports 82.52% for four-class decoding of BCI Competition IV-2a when training and test data come from one person, but 58.64% when the test person is unseen, a loss of 23.88 points [8]. Only the second number describes the experience of someone using the system for the first time.

Deep learning has reshaped the design of MI decoders. Convolutional networks can learn temporal and spatial filters jointly from raw signals [9], and EEGNet demonstrated that depthwise and separable convolutions achieve this with only a few thousand weights [10]. Newer models add self-attention on top: EEG Conformer follows a convolutional stem with a Transformer encoder [11], ATCNet runs multi-head attention within sliding time windows [12], and CTNet merges the two designs [8]. Surveys record steadily rising accuracy [13]–[15]. Most of these improvements, however, are measured within subject. The few cross-subject numbers come from protocols that use different numbers of training subjects, different analysis windows and epoch budgets, with or without alignment, and with different rules for deciding when to stop training. Results produced under such different conditions cannot be put into a single ranking [16]. Published figures are also easily quoted without their protocol, and within-subject results are often cited next to cross-subject ones as if they were comparable (Section VI-P).

This paper describes a controlled study built around one narrow question: on a small public dataset, does adding capacity—attention layers, depth or width—make a decoder better at people it has never seen? We planned the study before running experiments, fixing the dataset, the preprocessing, the leave-one-subject-out (LOSO) folds, a training recipe common to all deep models, the statistical tests and a set of decision rules: design choices are judged on a held-out validation subject, each test subject is scored once per configuration, one change is made per step, and negative results are reported. The plan did not survive unchanged—most importantly, we corrected the evaluation protocol midway—and we report every deviation. Within this framework we implemented five baselines and a hybrid, HCT-Net, which feeds the output of an EEGNet-like convolutional front end into a small windowed

Transformer encoder, and we added two Riemannian baselines that have no tuned hyperparameters.

We address four research questions. **RQ1:** Under one LOSO protocol, does cross-subject accuracy grow with model capacity? **RQ2:** Which components of a convolution–Transformer hybrid help on unseen users, and which do not? **RQ3:** How much do label-free adaptation steps that use only the new user's unlabelled trials contribute? **RQ4:** If capacity is not the limit, what is?

Our contributions are as follows.

- A comparison of six decoders (one classical, five deep, 3,188 to 697,412 parameters) and two Riemannian baselines under one LOSO fold structure, with a within-subject reproduction check for each deep baseline and a record of the protocols behind the published figures usually cited for them.
- Evidence, within the limits of one dataset and nine subjects, that extra capacity did not improve cross-subject decoding: under the first protocol accuracy was flat from 3,188 to 697,412 parameters, and under the corrected protocol it was flat across HCT-Net sizes from 3,924 to 24,836 parameters, while an 18-point gap between familiar and unseen users remained at every size.
- A component study in which Euclidean alignment was the only addition that consistently helped unseen users, while attention, the block the hybrid was designed around, did not.
- A correction to the evaluation protocol—early stopping and model selection on separate data—and a seed-noise analysis across all configurations, together with confidence intervals, a power analysis and a nested re-check of the final model selection.
- HCT-Net configuration A, a 3,924-parameter hybrid with label-free batch-normalisation adaptation that matches, but does not beat, EEGNet, released with data checksums, per-fold logs and code.
- An exploratory per-subject analysis that links cross-subject difficulty to how much each person's own signals change between recording sessions ($r = -0.89$, 95% CI -0.98 to -0.54).

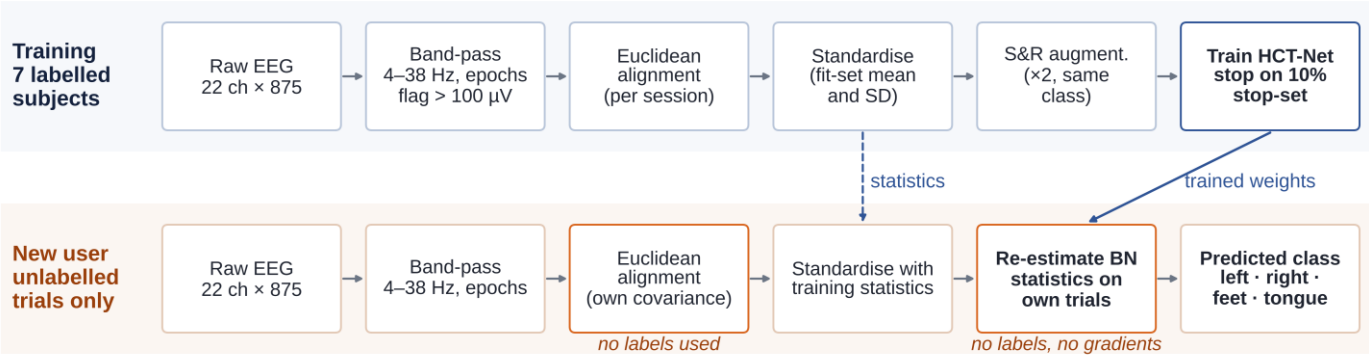


Fig. 1. System overview. Top: training on the labelled trials of seven subjects. Bottom: decoding a new user from unlabelled trials only. The two steps that adapt to the new user—Euclidean alignment from the user's own mean covariance and re-estimation of batch-normalisation statistics—use no labels and no gradient updates, so the system requires no calibration session.

Fig. 1 summarises the resulting system. The rest of the paper is organised as follows. Section II reviews related work. Section III describes the dataset and the exploratory analyses that shaped the pipeline. Section IV presents the preprocessing, the proposed model, the baselines and the adaptation methods, and Section V the evaluation protocol. Section VI reports the results, Section VII discusses them, Section VIII lists the limitations and Section IX concludes.

II. Background and Related Work

A. Motor Imagery and Its Neural Signature

Imagined and executed movements recruit largely overlapping cortical networks. Over the sensorimotor cortex, imagery shows up as a drop in mu- and beta-band power that starts soon after the cue and persists while the imagery continues, with a rebound once it ends [3]. For hand imagery the ERD is strongest over the opposite hemisphere (C3 for the right hand, C4 for the left); for the feet it is centred near the vertex (Cz), and tongue imagery yields a weaker and more diffuse pattern [2]. Decoding single trials is still hard, because the effect amounts to only a few decibels, volume conduction smears it over neighbouring channels, and its position and strength differ across people and across sessions [5].

B. Classical Decoding Pipelines

Classical pipelines compute band-power features after spatial filtering. Common spatial patterns (CSP) derive filters whose output variance differs as much as possible between two classes [17], [18]. Filter-bank CSP (FBCSP) repeats CSP in several frequency bands and keeps the most informative features according to mutual information [19]; it was the winning method on the four-class BCI Competition IV-2a [20]. Riemannian approaches instead classify the trial covariance matrices directly, exploiting the geometry of symmetric positive-definite matrices [21]. A review of a decade of work found that such methods remain hard to beat, especially when training data are scarce [15]. Our classical reference is FBCSP followed by linear discriminant analysis.

C. Convolutional Decoders

Schirrmeister *et al.* demonstrated that convolutional networks trained end to end on raw EEG can equal FBCSP, proposing the ShallowConvNet and DeepConvNet designs and releasing the braindecode library [9]. EEGNet [10] cut the number of parameters by about two orders of magnitude using three stages: temporal convolutions that act as learned frequency filters, depthwise convolutions that learn spatial

patterns for each of those filters, and a separable convolution that combines the resulting feature maps. Because it is so small, EEGNet is a natural baseline when data are limited, and it has served as the front end of several later hybrids, ours included.

D. Attention-Based Hybrids

The Transformer [22] has mostly been used for EEG as a second stage behind convolutional layers. EEG Conformer passes the output of a convolutional stem through self-attention layers and trains with segmentation-and-reconstruction augmentation [11]. ATCNet computes multi-head self-attention inside overlapping time windows and feeds each window to a temporal convolutional network, a design motivated by the brevity of MI trials [12]. CTNet keeps the convolution-then-Transformer order with a deeper encoder and reports both subject-specific and subject-independent accuracy [8]. All three are much larger than EEGNet: for our 875-sample input, the braindecode versions of ATCNet, CTNet and EEG Conformer contain 113,732, 152,364 and 697,412 trainable parameters, compared with 3,188 for EEGNet.

E. Cross-Subject Transfer

Jayaram *et al.* [6] surveyed transfer learning for BCIs, and Lotte [4] focused on methods that shorten calibration. One simple and effective strategy re-centres the data of every subject so that all subjects share the same average spatial covariance. In Riemannian re-centring, each subject's mean covariance is moved to the identity matrix on the manifold of covariance matrices [23]. Euclidean alignment (EA) achieves the equivalent in Euclidean space: each recording is whitened with the inverse square root of its arithmetic-mean covariance, so the aligned data can be passed to any classifier, deep networks included [24]. A large evaluation reported that EA benefits cross-subject deep learning across several datasets [25]. Since EA needs no labels, it can be computed for a new user from unlabelled recordings only. A second family learns features that do not reveal subject identity, for instance by training against a subject discriminator through gradient reversal [26]. A third family modifies the trained network at test time: AdaBN recomputes batch-normalisation statistics on the target data [27], TENT also tunes the normalisation parameters so as to minimise the entropy of the predictions [28], and Wimpff *et al.* studied such adaptation in an online MI setting [29]. We compare alignment, adversarial training and test-time adaptation within the same protocol.

F. Data Augmentation and Regularisation

With small EEG datasets, regularisation matters a great deal. Compact networks such as EEGNet restrict capacity by means of depthwise and separable convolutions, a max-norm limit on the spatial filters and dropout [10], [30]. Augmentation is the other usual tool. EEG Conformer and CTNet use segmentation and reconstruction (S&R): new trials are assembled from time segments cut out of different real trials of one class, so each synthetic example is made entirely of recorded signal [8], [11]. Label smoothing [31] and weight decay additionally penalise over-confident solutions. Whether such measures help a model generalise to new *people*, rather than to new trials from people already seen, is seldom tested on its own; our component study tests it for augmentation.

G. Reducing Calibration

Approaches to shorter calibration range from applying a decoder trained on other users with no target data at all, through unsupervised adaptation with unlabelled target data, to fine-tuning on a few labelled target trials [4], [6]. Decoders that need no calibration are the most convenient for users and the hardest to build, because nothing about the new user is known beyond what can be estimated without labels. Methods that use unlabelled target data—alignment and test-time adaptation—sit at the boundary: they need a short recording, but no instructions, cues or labels [24], [29]. This is the setting we study: the new user contributes unlabelled trials and nothing else.

H. Evaluation Practice

Three methodological issues recur in this literature. First, protocols differ: the MOABB initiative showed how strongly conclusions depend on the evaluation pipeline and argued for common benchmarks [16]. Second, the number of independent units is small—nine subjects in BCI IV-2a—so cross-validated estimates have wide error bars [32] and comparisons should use paired non-parametric tests over subjects with correction for multiple comparisons [33]. Third, information can leak from evaluation data into training through normalisation statistics, augmentation or model selection, a failure mode documented across many fields [34]. Table I collects the published figures for the decoders we compare, with the protocol that produced each one; only the CTNet study re-ran several baselines under a single subject-independent protocol [8].

TABLE I
PUBLISHED ACCURACY (%) FOR THE COMPARED DECODERS ON BCI IV-2A

Model	Params	Within	LOSO	Source and protocol
FBCSP	—	67.7[b]	—	Ang *et al.* [20]; $\kappa = 0.569$, session T → E
EEGNet	3,188	—	56.85[a]	Zhao *et al.* re-implementation [8]
ATCNet	113,732	85.38	—	Altaheri *et al.* [12]; subject-dependent
EEG Conformer	697,412	78.66	53.41[a]	Song *et al.* [11]; session split
CTNet	152,364	82.52	58.64	Zhao *et al.* [8]

Params: trainable parameters of the braindecode implementations at our 875-sample input. [a]Zhao *et al.*'s re-implementations (EEGNet+, EEG Conformer+) under their LOSO protocol: eight training subjects, 2–6 s window (1,000 samples), 600 fixed epochs, batch 512, no alignment. [b]Accuracy equivalent to the reported Cohen's κ for four balanced classes. ATCNet's paper also reports a subject-independent figure whose protocol we could not recover; it is omitted. No verified within-subject figure for EEGNet on this split was found (Section VI-P).

III. DATASET AND EXPLORATORY ANALYSIS

A. BCI Competition IV Dataset 2a

Our experiments use dataset 2a of BCI Competition IV [35], collected by the Graz BCI group [36] and now hosted by the BNCI Horizon 2020 repository (entry 001-2014) under a CC BY-ND 4.0 licence. Nine volunteers, coded A01 to A09, imagined four movements in response to visual cues: the left hand, the right hand, both feet and the tongue. The metadata inside the files list five men and four women aged between 17 and 26 years (mean 23.1). Twenty-two scalp electrodes covered the sensorimotor region (Fig. 2), and three further electrodes

recorded eye movements (EOG). The amplifier sampled at 250 Hz with a 0.5–100 Hz pass-band and a 50 Hz notch filter [36].

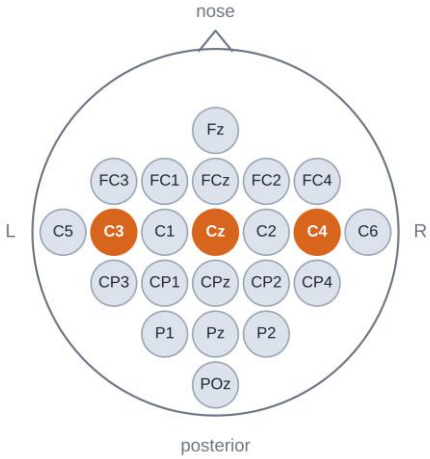


Fig. 2. Positions of the 22 EEG electrodes of BCI IV-2a on a schematic 10–20 map seen from above. C3 and C4 sit over the hand areas of the left and right sensorimotor cortex, and Cz lies near the vertex, over the foot area.

A trial opened at $t = 0$ s with a fixation cross displayed and a brief acoustic warning (Fig. 3). Two seconds later an arrow appeared for 1.25 s; its direction (left, right, down or up) told the participant which of the four movements to imagine, and the imagery was maintained until the cross vanished at $t = 6$ s, after which the screen went blank for a short pause. Every participant attended two sessions on separate days. A session consisted of six runs of 48 trials with 12 trials per class, so there are 288 trials per session, 576 per participant and 5,184 overall, in equal numbers for each class (Table II). We use both the first session (T) and the second session (E) of every participant; the labels of the evaluation session are part of the distributed files.

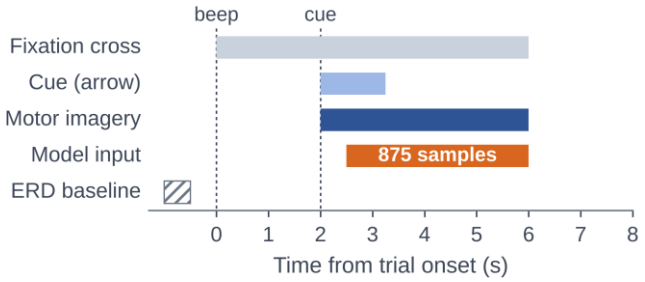


Fig. 3. Timeline of one BCI IV-2a trial. The network sees the 3.5 s segment between 0.5 and 4.0 s after cue onset (875 samples at 250 Hz), which leaves out the visual response evoked by the arrow itself. The hatched interval, from 3.0 to 2.5 s before the cue and therefore inside the preceding pause between trials, is the resting reference used only for the ERD analysis of Fig. 5.

TABLE II
DATASET AND ANALYSIS WINDOW

Property	Value
Participants	9 (A01–A09); 5 male, 4 female; 17–26 years
EEG channels	22 sensorimotor electrodes (plus 3 EOG, unused)
Sampling rate	250 Hz
Classes	left hand, right hand, feet, tongue
Sessions	2 per participant, on different days
Trials	288 per session (72 per class); 576 per participant
Total	5,184 labelled trials, class-balanced
Analysis window	0.5–4.0 s post-cue (875 samples)
Source	BNCI Horizon 001-2014 (.mat), 18 files, 744 MB

B. Data Preparation

Each of the 18 raw files (one per participant and session) was downloaded a single time and its SHA-256 checksum recorded, so that every number in this paper can be tied to the exact input bytes. From every run we retained the 22 EEG channels and filtered the continuous recording to 4–38 Hz with a fourth-order Butterworth design run in both directions, which cancels the phase shift. Some runs contain brief gaps of missing samples; we set these to zero before filtering and flagged every trial that overlapped a gap as an artefact. Epochs were then extracted between 0.5 s and 4.0 s after the cue appeared, 875 samples per channel, so that the visual response evoked by the arrow lies outside the window.

A trial was flagged, not deleted, if its peak-to-peak amplitude on any channel was above 100 µV. On average 4.3% of a participant's trials were flagged, but the rate varied widely (Table III): five participants lost under 1.1% of their trials, while A09 lost 25.0%. For A09 the flags fell mostly on one class—73 of 144 tongue trials, compared with 22–27 for each of the other three classes—and the largest amplitudes occurred at posterior channels (POz, CP4, CP3). A fixed threshold thus removed one class selectively for a participant with large signals, so we apply the rule only to the training subjects; validation and test subjects are always scored on all 576 of their trials, and no evaluation set loses half of a class.

TABLE III
TRIALS FLAGGED BY THE 100 µV PEAK-TO-PEAK RULE

Subject	Flagged	Left	Right	Feet	Tongue
A01	0.0%	0	0	0	0
A02	2.4%	7	1	4	2
A03	1.0%	1	2	0	3
A04	0.3%	0	0	0	2
A05	0.0%	0	0	0	0
A06	6.1%	7	8	11	9
A07	0.0%	0	0	0	0
A08	4.0%	0	2	2	19
A09	25.0%	22	22	27	73

Counts are out of 144 trials per class per subject (both sessions). Flagged trials are removed from training subjects only.

C. Exploratory Analysis

1) Spectral content: Inside the analysis window the spectra at C3, Cz and C4 peak clearly in the mu band, at 10–11 Hz, for

every class, with only a faint shoulder in the beta range (Fig. 4). What separates the classes is mainly the height of that peak. Averaged over 8–13 Hz, left-hand imagery gives the lowest power at all three electrodes and tongue imagery the highest, a difference of roughly 2 dB at C3 and C4 (at C3, 5.1 dB for the left hand versus 7.1 dB for the tongue). That differences of this size become visible only after averaging thousands of trials shows how weak the single-trial signal is that a decoder has to detect.

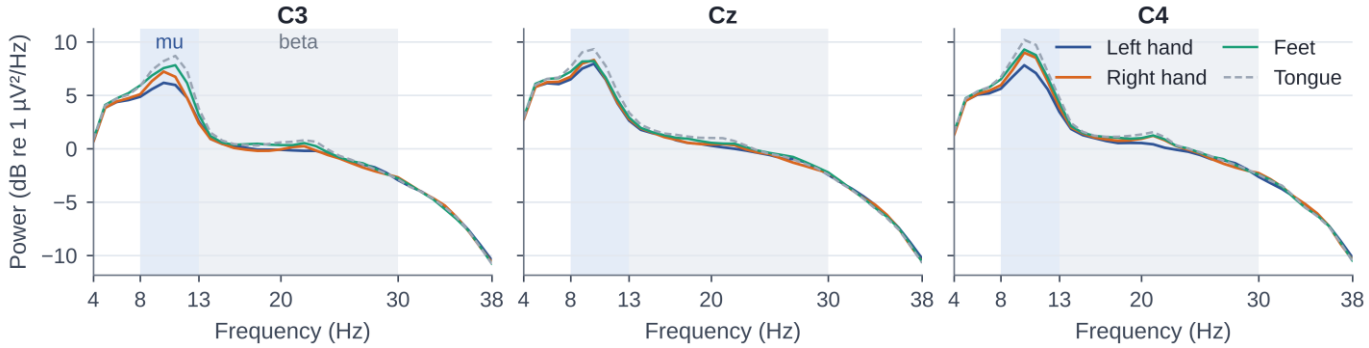


Fig. 4. Welch power spectral density (1 s segments) at C3, Cz and C4, computed over the analysis window (0.5–4.0 s following the cue) and averaged across artefact-free trials and the nine participants. Shading marks the 8–13 Hz mu range and the 13–30 Hz beta range. Every class peaks at 10–11 Hz in the mu band; mu power is lowest for left-hand and highest for tongue imagery.

2) Event-related desynchronisation: To check that the expected physiology is visible in these recordings, we cut longer epochs from 3 s before to 4 s after the cue and computed spectrograms at C3, Cz and C4 (128-sample windows overlapping by 96 samples), averaging first across artefact-free trials and then across participants. The reference period turned out to matter. The fixation interval before the cue is not a resting state: the acoustic warning that opens each trial already suppresses the mu rhythm, so using it as the reference made power after the cue seem to rise. We therefore took the pause between trials, 3.0 to 2.5 s before the cue, as the reference; it lies after the imagery of the previous trial has ended and before the warning of the current one. Relative to this reference all four classes show mu-band ERD after the cue (Fig. 5). Hand imagery suppressed the electrode on the opposite side: averaged from 0.5 s onwards, right-hand imagery reached −2.62 dB at C3 and left-hand imagery −2.58 dB at C4. Taking as onset the first point at which power fell 1 dB below the condition's own mean in the second preceding the cue, these effects began 0.33 s (right hand) and 0.46 s (left hand) after cue onset. Foot imagery was strongest at the vertex (−2.45 dB at Cz, against −1.64 and −1.47 dB at C3 and C4), and tongue imagery was the weakest at every electrode (between −0.87 and −1.71 dB).

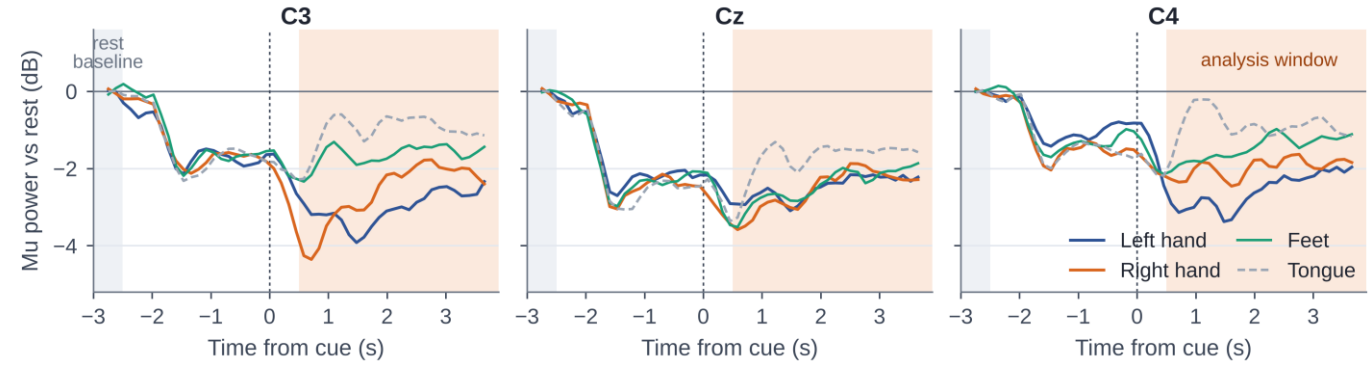


Fig. 5. Mu-band (8–13 Hz) power at C3, Cz and C4 relative to the resting reference taken in the pause between trials, averaged across artefact-free trials and the nine participants. The dashed line marks cue onset and the shaded band is the window the model sees. The dip around −2 s is caused by the acoustic warning that starts every trial. Hand imagery suppresses the opposite-side electrode (right hand at C3, left hand at C4), foot imagery is strongest at Cz and tongue imagery is weakest; C3 is also strongly suppressed during left-hand imagery.

The two hemispheres do not behave symmetrically. During left-hand imagery C3, on the same side as the hand, fell by 3.02 dB, more than the opposite electrode C4, whereas during right-hand imagery C4 fell by only 1.98 dB. In absolute terms the ERD is therefore largely bilateral with a bias towards the left hemisphere, and the lateralised component becomes clear only when the classes are compared: the ratio of C4 to C3 mu power was 1.649 dB for right-hand imagery and 1.043 dB for left-hand imagery (Mann–Whitney test over all 2,592 hand trials, $p = 9.3 \times 10^{-13}$). Decoders that learn spatial *patterns* over many electrodes should therefore have an advantage over features built from the power of single channels.

3) Subject identity in the signal: If every trial carries a strong imprint of the person who produced it, decoding across people is difficult. To measure this imprint we described each artefact-free trial by the log-variance of its 22 channels and counted how often a trial lies nearer to the centroid of its own subject than to that of any other. Without alignment the correct subject was found for 46.8% of trials, where chance is 11.1%; after Euclidean alignment of each subject (Section IV-B) the rate dropped to 22.9% (Fig. 6). Silhouette scores were negative in both cases (−0.076 and −0.089) because the nine clusters overlap strongly; since silhouette quantifies separation rather than identifiability, it does not answer this question. The

average log-Euclidean distance between the mean covariance matrices of different subjects dropped from 4.41 to practically zero, which is guaranteed, because alignment turns every mean covariance into the identity.

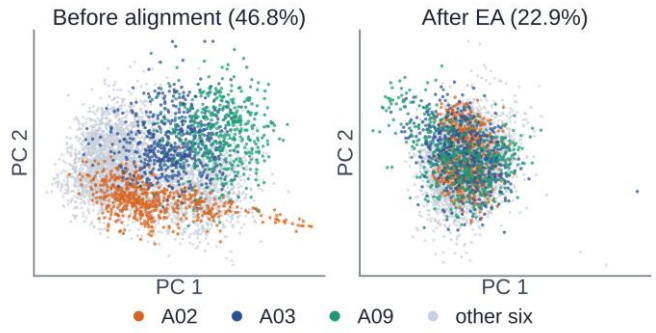


Fig. 6. Projection of the per-channel log-variance of every artefact-free trial onto its first two principal components, without and with Euclidean alignment; three participants are highlighted. The percentages are nearest-centroid subject recovery rates computed on all 22 features (chance 11.1%).

4) Session-to-session shift: Each participant's two sessions took place on different days. To quantify how much the recordings changed in between, we computed $\|\log R_T - \log R_E\|_F$, the log-Euclidean distance between the mean covariance matrices of the two sessions, using artefact-free trials before alignment. Values ranged from 1.03 for A01 to 3.38 for A02, with A05 second at 3.01. Alignment sets this distance to zero by design, yet Section VI shows that the participants whose sessions differ most remain the hardest to decode, which points to a part of each recording's identity that lies beyond second-order statistics and survives alignment.

IV. Methods

A. Problem Formulation

A trial is a matrix $X \in \mathbb{R}^{C\times T}$ with $C = 22$ channels and $T = 875$ samples, and its class is $y \in \{1, 2, 3, 4\}$. In each of the nine LOSO folds the subjects are split into seven for training, one for validation and one for testing (Section V). Only labelled trials of the training subjects are used to fit the decoder. At test time the decoder is given the test subject's trials without their labels. Two steps make use of these unlabelled trials, Euclidean alignment and the re-estimation of normalisation statistics, and both are computed from all of the subject's trials before the first prediction, as if the user had recorded an unlabelled warm-up block. Labels of the validation and test subjects serve only to score predictions (Fig. 1).

B. Euclidean Alignment, Standardisation and Augmentation

Euclidean alignment [24] whitens each recording using its own average spatial covariance. For a session s containing N_s trials,

$$R_s = \frac{1}{N_s}\sum_{i=1}^{N_s}\frac{1}{T}X_i X_i^{\top} \quad (1)$$

$$Z_i = {R_s}^{-1/2} X_i \quad (2)$$

where the inverse matrix square root is obtained from a symmetric eigendecomposition, with very small eigenvalues clipped. After this step every session has the identity as its mean covariance. Because no labels are involved, the same step can be applied to the unlabelled trials of a new user. Since the two sessions of a participant differ (Section III-C), we align each session on its own rather than each participant as a whole, and we treat training, validation and test subjects in exactly the same way.

Next, every channel is standardised using the mean μ_c and standard deviation σ_c computed on the trials that drive gradient updates (the *fit set*, Section V-C); these statistics are reused without change for all other splits. The fit set is then doubled with segmentation-and-reconstruction (S&R) augmentation [8], [11]. The time axis of each trial is divided into eight equal parts, and an artificial trial of class k is assembled part by part, taking its j-th part from a real class-k trial drawn at random. Artificial trials therefore consist only of recorded EEG and keep the classes balanced. No augmentation is applied to the stop-set or to validation or test data.

C. HCT-Net: A Hybrid Convolution–Transformer Network

HCT-Net (Fig. 7) connects a convolutional front end modelled on EEGNet [10] to a small Transformer encoder [22] that operates within sliding time windows, as in ATCNet [12]. Placing convolution before attention follows EEG Conformer and CTNet [8], [11]. None of the building blocks is new; what we test is whether this particular, compact combination generalises to people outside the training set. We added one constraint of our own: the number of convolutional feature maps equals the model dimension of the encoder, so the two halves connect without a projection layer and the parameter count is fully determined by two width settings.

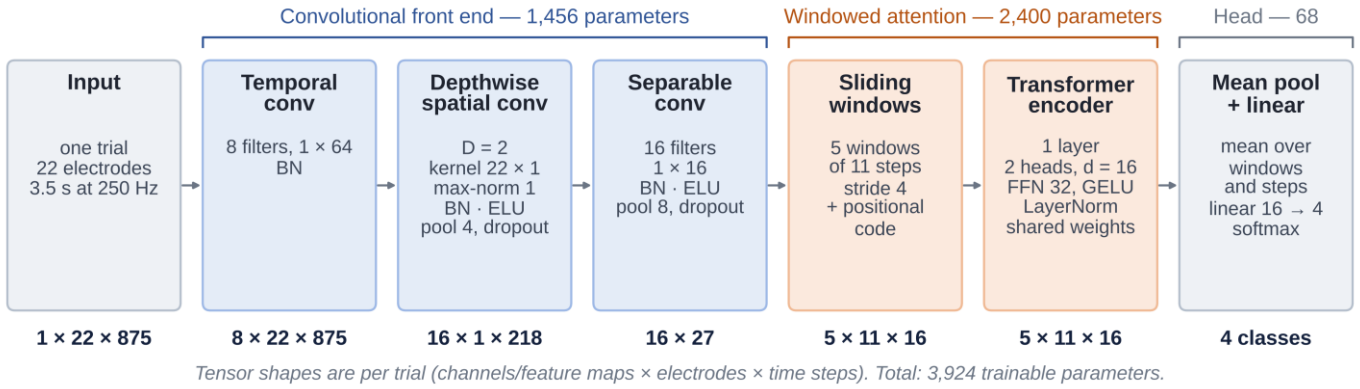


Fig. 7. HCT-Net configuration A. The convolutional front end, modelled on EEGNet, maps a 22 × 875 trial to 27 time steps with 16 features each. Five overlapping windows of 11 steps, each with a learned positional code added, are processed by the same one-layer, two-head Transformer encoder; the outputs are averaged over windows and time steps and passed to a linear classifier. Batch-normalisation (BN) layers appear only in the convolutional part.

1) Convolutional front end: The first layer applies F_1 temporal kernels of 64 samples (0.256 s), which behave like a set of learned frequency filters. A depthwise convolution spanning all 22 electrodes then derives $D = 2$ spatial filters from each temporal filter; their weight norms are limited to 1.0 (a max-norm constraint) so that individual channels cannot dominate. This is followed by batch normalisation [37], an ELU activation [38], average pooling with factor 4 and dropout at rate 0.25 [30]. A separable convolution—a depthwise kernel of 16 samples and a pointwise combination into F_2 feature maps—is again followed by batch normalisation, ELU, pooling with factor 8 and dropout, which leaves 27 time steps of F_2 features from the original 875 samples. The convolutions carry no bias terms, as each feeds directly into batch normalisation.

2) Windowed attention: Write $z_t \in \mathbb{R}^d$, with $d = F_2$, for the feature vector at pooled step t. The sequence of 27 steps is divided into $W = 5$ windows of $L = 11$ steps that start 4 steps apart, and a learnable positional code $P \in \mathbb{R}^{L\times d}$, common to all windows, is added to each:

$$H^{(w)} = [z_{4w+1}, \dots, z_{4w+11}]^{\top} + P, \quad w = 0, \dots, 4 \qquad (3)$$

All windows share one Transformer encoder. In the final model it has a single layer with two attention heads, model dimension d, a feed-forward layer of width $2d$ with GELU activation [39], dropout of 0.25 and layer normalisation after each sub-layer [40]. Inside a window each head applies scaled dot-product attention across the 11 time steps. Queries, keys and values are linear projections of the window, $Q = HW_Q$, $K = HW_K$ and $V = HW_V$, and each head has dimension $d_h = d/2$ [22]:

$$Attention(Q,K,V) = softmax\left(\frac{QK^{\top}}{\sqrt{d_h}}\right)V \qquad (4)$$

The outputs of the two heads are concatenated and mapped back to d dimensions before the feed-forward block; both sub-layers use a residual connection followed by layer normalisation. The encoder outputs $E^{(w)} \in \mathbb{R}^{L\times d}$ are averaged over all windows and time steps, and a linear layer turns the result into class probabilities:

$$f = \frac{1}{WL}\sum_{w=0}^{W-1}\sum_{t=1}^{L} E^{(w)}{}_t, \quad p = softmax(Wf + b) \qquad (5)$$

For configuration A ($F_1 = 8$, $F_2 = d = 16$, one encoder layer) the parameters split into 1,456 in the convolutional front end, 2,224 in the encoder, 176 in the positional code and 68 in the classifier, 3,924 in all, close to EEGNet's 3,188. The configuration written into our design before any experiment ($F_1 = 16$, $F_2 = 32$, two encoder layers) has 20,996 parameters, 81% of them in the encoder; we call it configuration D. Table IV details the layers and Table V the four sizes compared in the size sweep.

TABLE IV
HCT-NET LAYERS, OUTPUT SHAPES AND PARAMETERS

Block	Operation	Output (A)	A	D
Input	one trial, 22 × 875	1×22×875	—	—
Temporal	F_1 conv 1×64, BN	8×22×875	528	1,056
Spatial	depthwise 22×1, $D = 2$, max-norm 1; BN, ELU, pool 4, dropout	16×1×218	384	768
Separable	depthwise 1×16 + pointwise to F_2; BN, ELU, pool 8, dropout	16×27	544	1,600
Windows	5 × 11 steps, stride 4; positional code	5×11×16	176	352
Encoder	Transformer layer(s), 2 heads, FFN $2d$	5×11×16	2,224	17,088
Head	mean over windows and steps; linear	4	68	132
Total			3,924	20,996

A: $F_1 = 8$, $F_2 = d = 16$, one encoder layer (final model). D: $F_1 = 16$, $F_2 = d = 32$, two layers (initial design). Shapes are per trial.

TABLE V
SIZE CONFIGURATIONS OF HCT-NET

Config	F_1	$F_2 = d$	Layers	Conv	Encoder	Pos.	Head	Total
A	8	16	1	1,456	2,224	176	68	3,924
B	8	16	2	1,456	4,448	176	68	6,148
C	16	32	1	3,424	8,544	352	132	12,452
D	16	32	2	3,424	17,088	352	132	20,996

The feed-forward width is $2d$ in every configuration, which reproduces the predicted parameter counts exactly.

D. Baselines

We re-implemented **FBCSP + LDA** [19] following its original description. Nine sub-bands of 4 Hz width span 4–38 Hz (the last covering 36–38 Hz), and each is obtained with a fourth-order Butterworth filter. Within every band and for every one-versus-rest split of class k, CSP solves the generalised eigenvalue problem

$$\Sigma_k w = \lambda(\Sigma_k + \Sigma_{\neg k})w \quad (6)$$

in which Σ_k and $\Sigma_{\neg k}$ denote the mean trial covariance of class k and of the remaining classes. For each class and band we retain the two eigenvectors with the largest and the two with the smallest eigenvalues, and represent a trial by the log-variance of its spatially filtered signals. A mutual-information ranking keeps 16 features—matching the reference algorithm, which uses four one-versus-rest classifiers with four features each—and a linear discriminant classifier with shrinkage predicts the class. Spatial filters, feature selection and classifier are all fitted on training trials alone.

Riemannian baselines. Two further classical references work directly on trial covariance matrices (22 × 22, estimated with oracle-approximating shrinkage). Minimum distance to mean (MDM) assigns a trial to the class whose Riemannian mean covariance is closest under the affine-invariant metric, and the tangent-space classifier (TS-LR) projects covariances onto the tangent space at their mean and applies logistic regression with $C = 1$ [21]. MDM is run with and without per-session Riemannian re-centring (RC), which whitens every session by the Riemannian mean of its own covariances [23], the Riemannian counterpart of Euclidean alignment; TS-LR is run with re-centring. Both use the same preprocessing, artefact rule and seven training subjects per fold as the deep models. They are deterministic and have no setting chosen on validation data, so they need neither seeds nor a selection subject. They were added after the main study, using pyRiemann 0.12.

For **EEGNet, ATCNet, CTNet and EEG Conformer** we use the reference implementations in braindecode 1.8.1 [9] with default hyperparameters, setting only the input shape (22 × 875) and the number of classes. Relying on the published architectures instead of our own re-implementations ensures that any shortfall against a published result reflects the protocol and not a coding error on our side. EEGNet is used in its EEGNet-8,2 form ($F_1 = 8$ temporal filters, $D = 2$, $F_2 = 16$), with a dense classifier over all 27 pooled time steps that accounts for 1,732 of its 3,188 parameters. ATCNet joins an EEGNet-like convolutional block to five sliding windows, each handled by multi-head self-attention and a two-layer temporal convolutional network. EEG Conformer stacks six Transformer encoder layers on a convolutional patch embedding, and CTNet places six encoder layers after a convolutional block. With our input length the four models have 3,188 (EEGNet), 113,732 (ATCNet), 152,364 (CTNet) and 697,412 (EEG Conformer) trainable parameters. Every deep model, HCT-Net included, gets the same preprocessing and the same training recipe, described next.

E. Training

Every deep model is trained to minimise the cross-entropy between its predicted distribution p and a label-smoothed target q [31]:

$$\mathcal{L} = -\sum_{k=1}^{4} q_k \log p_k, \quad q_k = (1-\varepsilon)\mathbb{1}[k = y] + \frac{\varepsilon}{4} \quad (7)$$

with $\varepsilon = 0.1$. Optimisation uses Adam [41] at a learning rate of 10^{-3} with weight decay 10^{-4}, cosine annealing of the learning rate over at most 500 epochs [42], batches of 64 trials, and early stopping that restores the best checkpoint once the accuracy on a held-out stop-set has not improved for 50 epochs (Section V-C). These settings were fixed in the design and are common to all models; none of the models was tuned individually (Appendix C).

F. Label-Free Test-Time Adaptation

After training on seven subjects, a network holds the feature statistics of those subjects in its batch-normalisation (BN) layers, each of which computes

$$BN(h) = \gamma \frac{h-\mu}{\sqrt{\sigma^2+\epsilon}} + \beta \quad (8)$$

where the running statistics μ and σ^2 are accumulated during training. In the spirit of AdaBN [27], which corresponds to TENT [28] without gradient updates, we discard these running statistics and replace them with the exact mean and variance over all unlabelled trials of the new subject (cumulative averaging with momentum switched off). All learned weights, γ and β included, stay as they are. The step uses neither labels nor gradients and takes about 0.2 s per subject. In configuration A it affects only the three BN layers of the convolutional front end, since the layer normalisation in the encoder keeps no running statistics.

Three alternatives to this *bn* variant were tested, each with settings chosen before the run. Full TENT also optimises the BN scale and shift parameters for one pass so that the prediction entropy $H(p) = -\sum_k p_k \log p_k$ decreases (Adam, learning rate 10^{-3}). Pseudo-label self-training fine-tunes the entire network for ten epochs (Adam, 10^{-4}) on those of the subject's trials whose predicted class has a confidence of at least 0.8, or at least 0.9, and keeps the unadapted model if fewer than 16 trials pass. All variants begin from an identical copy of the trained weights and are run separately on the validation and the test subject, which makes the comparison between variants exactly paired.

G. Subject-Adversarial Variant

We also trained a version of configuration D with an extra output head that tries to identify which training subject a trial came from. It reads the same features as the class head and is connected to them through a gradient reversal layer [26], which passes features through unchanged in the forward pass and multiplies the gradient by $-\lambda$ in the backward pass. This drives the feature extractor towards representations that hide the identity of the subject. The weight λ rises from zero during training and its final value is chosen per fold on the validation subject from 0.05, 0.2 and 0.5. Because this means three training runs per fold, the variant used a shorter budget than the other models: at most 200 epochs, with early stopping after 40 epochs without improvement.

V. Evaluation Protocol

A. Within-Subject Reproduction

Before running any cross-subject experiment we trained every baseline on session T and tested it on session E of the same participant, to compare our implementations with published within-subject results. According to the design, at least two baselines had to come within roughly three points of their published accuracy before work on the proposed model could begin. This check revealed a flaw in our first protocol. Reserving 20% of session T for early stopping left about 230 trials for training and only 58 for validation, so validation accuracy moved in steps of 1.7 points, too coarse to select a checkpoint, and losing a fifth of the training data hurt the

weakest participants most: EEGNet on A05 improved from 26.7% to 56.6% once it was trained on all 288 trials. The final within-subject protocol therefore trains on all session-T trials under a fixed 500-epoch schedule annealed to the end, and evaluates session E a single time. The diagnostics that led to this protocol, and to the FBCSP feature budget (Appendix B), were themselves run on session E, so these two choices were made after seeing within-subject test results. They affect only the within-subject reproduction figures; no cross-subject choice used test data.

B. Leave-One-Subject-Out Folds

Cross-subject accuracy is estimated with nine LOSO folds (Fig. 8a). Fold k tests on subject k, validates on subject $k + 1$ (A01 for fold 9) and trains on the other seven subjects. Each subject therefore serves exactly once as the test subject and exactly once as the validation subject. Setting a validation subject aside removes one eighth of the training data relative to the eight-subject protocol of Zhao *et al.* [8]; Section VI-G measures what this costs.

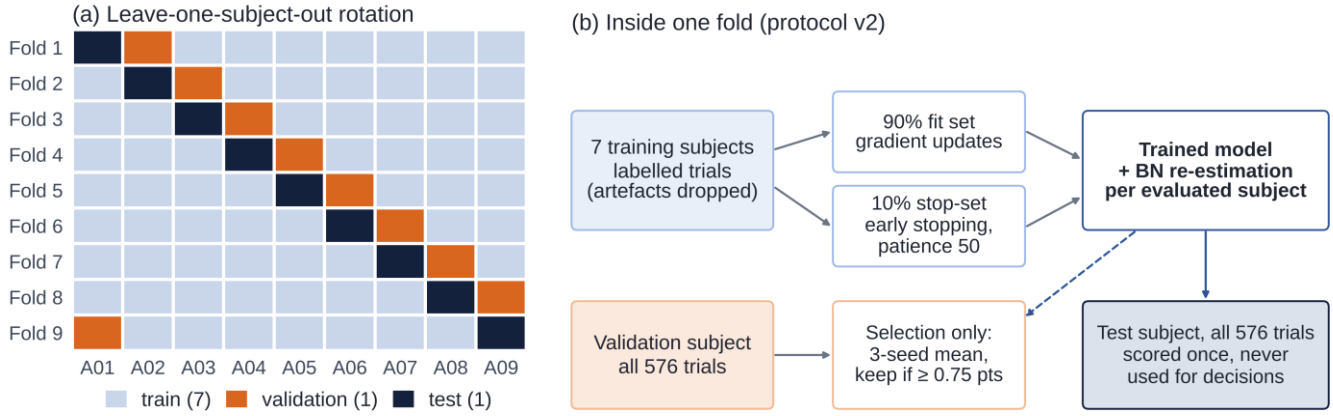


Fig. 8. The evaluation protocol. (a) Roles of the subjects in the nine LOSO folds. (b) What happens inside one fold under protocol v2: a 10% stop-set taken from the training subjects controls early stopping, the validation subject serves only to choose between configurations (mean of three seeds, 0.75-point threshold), and the test subject is scored a single time and never influences a decision.

C. Protocol v1 and Protocol v2

In the first protocol version (v1), which we used for the baseline comparison, the component study and the first round of model changes, the validation subject had two roles: it determined the early-stopping epoch and it decided which configuration to keep. Because a model whose training is stopped on a subject has, in effect, been tuned to that subject, validation accuracy came out higher than test accuracy. Protocol v2 gives the two roles to different data. Early stopping is driven by a stop-set made of a tenth of the training subjects' trials, stratified by subject and class and never used for gradient updates; the other 90% form the fit set, which also supplies the standardisation statistics. The validation subject plays no part in training or stopping and is used for selection only. Throughout the paper, results obtained under the two protocols are labelled v1 and v2 and never mixed.

D. Decision Rules

Each change to the model or the pipeline was judged by rules written down before it was measured. Rules 3 and 4 were added together with protocol v2 (Section V-C); before that, under v1, a change was kept if it raised single-seed validation accuracy at all (Section VI-H).

1) Only the validation subject informs decisions; test accuracy is logged but not looked at when deciding.
2) Changes are made individually, each on top of the most recently accepted configuration.
3) Every configuration is trained with three seeds (0, 1, 2); the seed sets the initial weights, the random augmentation and the split of the stop-set.
4) A change is kept only when it increases the validation accuracy averaged over three seeds by 0.75 points or more; if two options lie within that margin, the smaller model wins.
5) Negative results are logged with their date and values, and we disclose every case in which a previous test result affected the order in which experiments were run.

We set the 0.75-point threshold from the first noise measurement, made on one configuration without adaptation, in which three seeds moved the nine-fold validation mean by 0.9 points (standard deviation 0.46), so that the threshold was about two standard deviations of the difference between two three-seed means. Measured later across all nine three-seed configurations, the seed noise was about twice as large (Section VI-G); the threshold is therefore closer to one standard deviation of such a difference than to two, and decisions that cleared it by a small margin are correspondingly less certain.

E. Metrics and Statistical Testing

Performance is reported as classification accuracy (chance level 25% for four balanced classes) and as Cohen's kappa [43], $\kappa = (p_o - p_e)/(1 - p_e)$, in which p_o is the observed and p_e the chance agreement. We give results for each subject and as the mean over the nine subjects together with the sample standard deviation across subjects. To compare two models we apply the two-sided Wilcoxon signed-rank test to the nine pairs of per-subject accuracies [33], [44], with p-values taken from the exact null distribution as implemented in SciPy [45]; where some absolute differences are tied, they receive average ranks and this distribution is then a close approximation. When a model is compared with several others, the resulting p-values are adjusted with Holm's step-down procedure [46], and we report

the adjusted and the unadjusted values. Many tests in this paper fall outside any such family; their p-values are uncorrected and should be read as descriptive. Table XVIII additionally applies Holm correction across the eight comparisons on which our main conclusions rest. With nine pairs the lowest possible two-sided p-value is $2/2^9 \approx 0.0039$, which requires all nine subjects to change in the same direction; a single subject going the other way raises the floor to 0.0078. Effect sizes are given as the mean paired difference together with the number of subjects that improve.

F. Leakage Controls

Information about the evaluation subjects can enter a model through unexpected routes [34]. In our pipeline the normalisation statistics are computed on the fit set alone, augmentation touches only the fit set, artefact rejection removes trials only from training subjects, and the validation and test subjects supply unlabelled trials to just two label-free steps—Euclidean alignment and the re-estimation of batch-normalisation statistics—both of which use information a real user could provide before any labelling. Two automatic checks confirm that the mean covariance after alignment is the identity and that no artificial trial mixes segments from both sides of a fold boundary.

G. Implementation and Reproducibility

The software is written in Python 3.14 using PyTorch 2.14 with CUDA 12.6 [47], together with braindecode 1.8.1, SciPy [45], NumPy [48], scikit-learn [49] and, for the Riemannian baselines, pyRiemann 0.12, and the deep-learning experiments ran on one laptop with an NVIDIA RTX 4050 GPU (6 GB); FBCSP and the Riemannian baselines ran on the CPU. For every fold and seed a run stores one row with the accuracy, kappa, the epoch selected by early stopping, the training duration, the random seed and the code revision, and its configuration file is kept alongside; a reporting script in the repository rebuilds the project's result tables and significance tests from these logs in about one second, needing neither a GPU nor the raw data, and the additional analyses in this paper (intervals, power, chance levels and the Riemannian baselines) have scripts of their own in the repository's paper/ folder. Training on the GPU is not bit-for-bit deterministic. When a configuration was repeated with the same seed, most folds gave identical results, but a few stopped at a different epoch, which changed the nine-fold validation mean by no more than 0.63 points, inside the measured noise floor. The code, the raw data with checksums, the per-fold logs and the configuration files are published together with this paper.

VI. Results

We report the experiments in the order in which they were run, because several later decisions depended on earlier findings. The experiments up to and including Section VI-F use protocol v1 with one seed; those from Section VI-G onwards use protocol v2 with three seeds. Unless stated otherwise, accuracies are cross-subject test accuracies averaged over the nine LOSO folds.

A. Within-Subject Reproduction of the Baselines

Table VI sets our within-subject accuracies beside the published ones. EEG Conformer was 1.4 points short of its published accuracy and FBCSP 3.0 points short of the accuracy implied by its published κ, which met the reproduction criterion we had set in advance. ATCNet and CTNet, by contrast, fell 18.3 points below the figures available to us, and we examined every explanation we could think of. Both networks are the authors' reference implementations, run with default settings. Extending training from 500 to 1,500 epochs lowered accuracy slightly for both (on the three most decodable subjects ATCNet went from 80.0% to 77.6% and CTNet from 80.8% to 80.4%). A longer input of 1,125 samples, the length used in the ATCNet repository, gave no improvement on the weakest subjects (41.1%, compared with 43.9% at our window length in the same diagnostic run). The shortfall comes entirely from the hard subjects. On A01, A03 and A08, ATCNet reached 80.0% and CTNet 80.8%, close to the published means of 81.1% and 82.5%, but on A04, A05 and A06 ATCNet obtained only 49.3%, 43.4% and 39.6%. In our pipeline both models therefore reproduce on subjects with a clear signal and break down on difficult ones. HCT-Net in its initial design (configuration D) reached 73.2% within subject, the second-best of the six models and 7.3 points above EEGNet.

TABLE VI
Within-subject accuracy (%), session T → E

Model	Ours	Published	Gap
EEG Conformer	77.3	78.66 [11]	−1.4
HCT-Net (D)	73.2	—	—
EEGNet	65.9	—[a]	—
FBCSP + LDA	64.7	67.7[b] [20]	−3.0
CTNet	64.2	82.52 [8]	−18.3
ATCNet	62.8	81.10[c]	−18.3

Training used every session-T trial and a fixed schedule of 500 epochs; gaps are computed from the values shown. [a]No verified figure for this split (Section VI-P). [b]Ang *et al.* report Cohen's κ = 0.569 for one-versus-rest FBCSP; 67.7% is the equivalent accuracy for four balanced classes (0.75κ + 0.25). [c]Authors' repository, train–validation–test script; the paper reports 85.38% subject-dependent [12].

These numbers were obtained only after we had corrected the within-subject protocol itself (Section V-A); Table VII shows how much that mattered. When 20% of session T was set aside for early stopping, EEGNet reached 55.5% within subject, ATCNet 60.1%, EEG Conformer 74.4% and CTNet 52.9%. Using every session-T trial with a fixed schedule annealed to the end lifted these values by between 2.7 and 11.3 points. A diagnostic run with EEGNet on four subjects identified the cause: keeping the last annealed model rather than the best checkpoint made little difference, whereas giving back the fifth of the training data that had been held out lifted A05 from 26.7% to 56.6%. The participants with the weakest signals suffered most from the missing data (Appendix B).

TABLE VII
WITHIN-SUBJECT ACCURACY (%) BEFORE AND AFTER THE PROTOCOL FIX

Model	80/20 holdout	All of T	Gain
EEGNet	55.5	65.9	+10.4
ATCNet	60.1	62.8	+2.7
EEG Conformer	74.4	77.3	+2.9
CTNet	52.9	64.2	+11.3

80/20 holdout: early stopping on 20% of session T, best checkpoint restored. All of T: all 288 session-T trials, fixed 500-epoch cosine schedule, final model. The 80/20 runs are recorded in the project's findings log; the per-fold result files keep only the final protocol.

B. Cross-Subject Comparison Under One Protocol

Table VIII lists the cross-subject accuracy of every model on every test subject under protocol v1. ATCNet (52.4%), EEG Conformer (52.1%) and EEGNet (51.2%), the three strongest deep baselines, lay within 1.2 points of one another, ahead of CTNet (50.2%), HCT-Net D (46.9%) and FBCSP (44.2%). EEGNet was 7.0 points ahead of FBCSP (Wilcoxon $p = 0.039$, uncorrected); with one comparison among the many possible pairs of baselines, this difference would not survive correction for multiple testing. Differences between subjects dwarfed those between models: on A01 and A03 every deep model exceeded 60%, while on A02 none reached 32%.

TABLE VIII
CROSS-SUBJECT ACCURACY (%) PER TEST SUBJECT, PROTOCOL V1, ONE SEED

Model	Params	A01	A02	A03	A04	A05	A06	A07	A08	A09	Mean ± SD	κ
ATCNet	113,732	75.3	30.9	78.8	42.5	37.8	38.7	43.4	62.7	61.5	52.4 ± 17.5	0.365
EEG Conformer	697,412	71.9	31.6	67.2	41.0	42.2	42.0	45.1	65.3	63.0	52.1 ± 14.6	0.362
EEGNet	3,188	70.8	28.3	73.8	35.9	37.8	44.6	40.8	65.5	63.0	51.2 ± 17.0	0.349
CTNet	152,364	71.9	31.9	62.7	41.1	35.2	42.2	49.1	60.9	56.8	50.2 ± 13.6	0.336
HCT-Net D	20,996	72.7	28.3	70.1	34.7	34.4	33.0	34.9	62.0	52.1	46.9 ± 17.5	0.292
FBCSP + LDA	—	54.3	29.2	60.9	35.2	29.3	32.3	43.2	55.9	57.3	44.2 ± 13.0	0.256

SD: sample standard deviation across the nine test subjects. Euclidean alignment on for all models; early stopping on the validation subject (protocol v1).

HCT-Net in its initial design (configuration D) was second to last. Table IX lists the paired comparisons with Holm correction over the five baselines, as the design required: HCT-Net was 5.5 points below ATCNet ($p = 0.0039$, corrected 0.020), the only difference to survive correction, and 4.3 to 5.2 points below EEGNet and EEG Conformer at uncorrected $p < 0.05$. It lost most on the middle of the subject distribution—A06 (33.0% against EEGNet's 44.6%), A07 (34.9% against 40.8%) and A09 (52.1% against 63.0%)—while remaining competitive on the strongest subjects.

TABLE IX
HCT-NET D AGAINST EACH BASELINE, PROTOCOL V1

Against	Mean diff.	p	p (Holm)
ATCNet	−5.5	0.0039	0.020
EEG Conformer	−5.2	0.020	0.078
EEGNet	−4.3	0.023	0.078
CTNet	−3.3	0.21	0.42
FBCSP + LDA	+2.7	0.50	0.50

Wilcoxon signed-rank test over the nine subjects, two-sided; Holm–Bonferroni over the five comparisons.

C. Within-Subject Skill Did Not Predict Transfer

Fig. 9 sets each model's within-subject accuracy against its cross-subject accuracy. The drop from one to the other ranged from 10.4 points (ATCNet) to 26.3 points (HCT-Net D), the largest of all and larger than the 23.88-point drop reported for CTNet [8] that motivated this project. FBCSP dropped by 20.5 points, so the phenomenon is not specific to deep models. The model designed to close the gap had the largest gap: it was a good within-subject decoder that transferred worst.

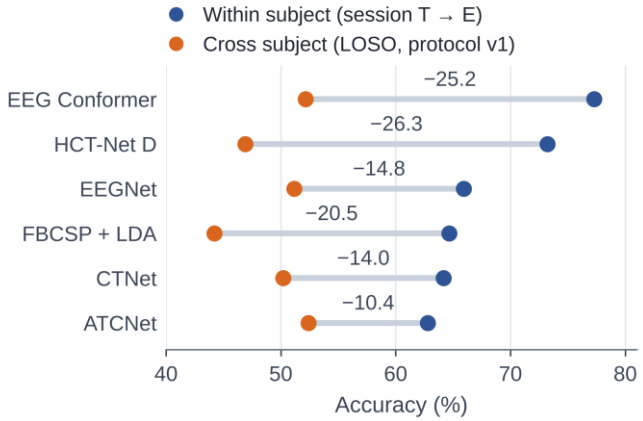


Fig. 9. Within-subject (session T → E) and cross-subject (LOSO, protocol v1) accuracy for the six models. Numbers give the drop in points. The two orderings disagree; six models are too few to estimate how strongly they are related.

Across the six models, within-subject accuracy did not predict cross-subject accuracy (Pearson $r = 0.08$, 95% CI −0.78 to 0.84, $p = 0.88$; Spearman $\rho = -0.14$). Six models are far too few to rule out a moderate relationship, so this is an absence of evidence that within-subject rankings carry over, not evidence that they cannot. An earlier analysis of ours correlated within-subject accuracy with the *drop* and obtained $r = 0.86$ ($p = 0.03$); that correlation is an artefact, because the drop is computed from within-subject accuracy and is therefore partly correlated with itself. We report it as a caution: on this evidence, a model that decodes familiar subjects better has not been shown to decode unseen subjects better, and within-subject benchmarks should not be used as a proxy for calibration-free performance.

D. Capacity Across Architectures and Depths

Fig. 10 relates cross-subject accuracy to the number of trainable parameters. Under protocol v1 (one seed, no adaptation) the four strongest deep models were separated by only 2.2 points (50.2–52.4%), although their sizes differ by

more than a factor of one hundred: EEG Conformer has 219 times as many parameters as EEGNet and was about one point more accurate, and ATCNet has 36 times as many for 1.2 points. With four models, the correlation between accuracy and the logarithm of the parameter count (r = 0.28, 95% CI −0.93 to 0.98) carries almost no information; what the data do show is the absence of any large gain. Changing the depth of HCT-Net D gave the same picture: with one, two and six encoder layers (12,452, 20,996 and 55,172 parameters) it reached 46.0%, 46.9% and 47.7%, and none of the pairwise differences was significant (L1–L2 p = 0.25, L2–L6 p = 0.50, L1–L6 p = 0.16). Under protocol v2, with three seeds and adaptation, the six HCT-Net configurations from 3,924 to 24,836 parameters lay within 1.9 points of each other on test (52.5–54.4%; the failed variant with batch normalisation in the encoder is excluded). The large baselines were not re-run under v2, so the comparison above 25,000 parameters rests on protocol v1 only.

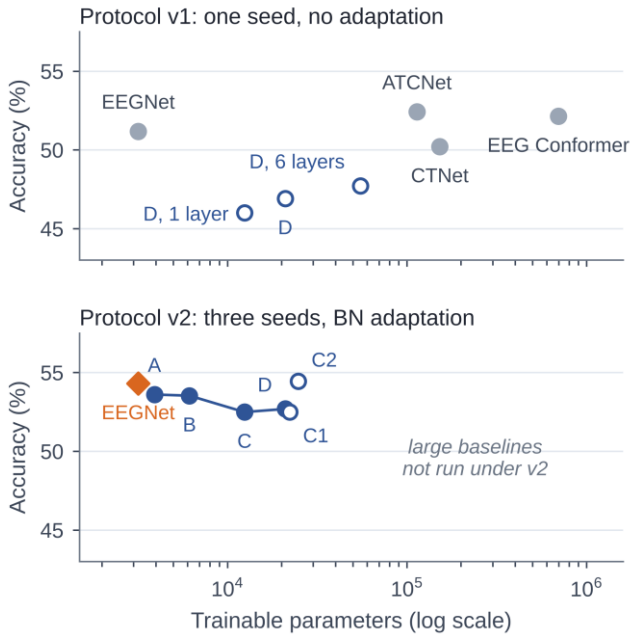


Fig. 10. Cross-subject test accuracy versus the number of trainable parameters (logarithmic axis), shown separately for the two protocols. Top: protocol v1, one seed, no test-time adaptation (open circles: configuration D with one, two and six encoder layers). Bottom: protocol v2, three-seed means with batch-normalisation adaptation (A–D: the HCT-Net sizes of Table V; C1, C2: the refinement variants of Table XIV). Within each protocol, accuracy does not rise with model size; the large baselines were run only under v1.

E. Component Study

To find out which parts of HCT-Net D produce its cross-subject accuracy, we removed one component at a time and re-ran the complete LOSO evaluation (Table X, Fig. 11). Of the three additions made to the convolutional core, one helped, one hurt and one had no measurable effect. Without Euclidean alignment (V3) accuracy fell by 5.1 points and seven of the nine subjects got worse; this was the largest single effect and similar to the 5.5 points that alignment added for FBCSP (Section VI-F). Taking out the attention encoder (V1) *raised* accuracy by 1.9 points, for eight of the nine subjects, and letting attention span the whole sequence of 27 steps instead of separate windows (V2) raised it by 2.9 points. Dropping augmentation (V4) changed accuracy by +1.2 points (p = 0.91), and keeping nothing but the convolutional part (V5) lowered it by 7.8 points.

This study was run under protocol v1, with one seed, on configuration D. It was not repeated for the final configuration A under protocol v2; in particular, configuration A without its encoder was never tested, so whether the attention layer of the final model contributes anything is not known.

TABLE X
COMPONENT STUDY OF HCT-NET D, PROTOCOL V1, ONE SEED

Variant	Acc.	vs full	Up	p	Holm
Full model (D)	46.9	—	—	—	—
V1 no attention	48.8	+1.9	8/9	0.098	0.29
V2 global attention	49.8	+2.9	5/9	0.16	0.33
V3 no alignment	41.8	−5.1	2/9	0.027	0.14
V4 no augmentation	48.1	+1.2	3/9	0.91	0.91
V5 convolution only	39.1	−7.8	2/9	0.027	0.14
Adversarial (GRL)	48.8	+1.9	7/9	0.098	—

Up: subjects on which the variant beats the full model. Holm correction over the five variants. The adversarial row adds a gradient-reversal subject classifier to the full model.

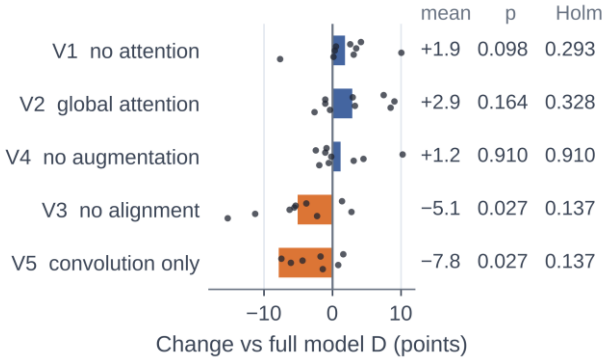


Fig. 11. Change in cross-subject accuracy when one component is removed from HCT-Net D (full model, 46.9%; protocol v1, one seed). Bars: means; dots: the nine subjects. Right: uncorrected and Holm-corrected Wilcoxon p-values. Alignment is the only addition whose removal consistently hurts.

No single effect survived Holm correction over the five variants; with nine subjects and two reversals the two-sided p-value cannot fall below 0.02, whatever the size of the effect. The directions are nevertheless consistent across experiments, and they contradicted two constraints fixed in the design. The locality prior behind windowed attention was not supported: global attention was better. And the subject-adversarial variant (Section IV-G) gained exactly as much as simply removing attention (48.8%, +1.9 points over the full model, p = 0.098), even though it was trained on a shorter budget, so it offered no benefit beyond V1. A layer-by-layer comparison also showed that V1 was not, as we had assumed, "EEGNet plus alignment": its classifier received 32 numbers after global average pooling, against EEGNet's 432 (27 steps × 16 filters), and its convolutional widths were twice EEGNet's. These two unintended differences account for the remaining 2.4 points between V1 and EEGNet.

F. The Effect of Euclidean Alignment

Alignment improved every model on which we tested it (Table XI). With FBCSP under protocol v1 it lifted cross-subject accuracy from 38.7% to 44.2% (+5.5 points) and helped seven of the nine subjects, gaining much on A08 (+20.1) and A07 (+15.6) and losing a little on A04 (−2.6) and A06 (−0.7); because of these two reversals the Wilcoxon test gives only p =

0.074. With EEGNet under protocol v2, test accuracy rose from 42.5% to 51.9% (+9.4 points) and all nine subjects gained (p = 0.0039). Estimating the reference covariance of each session with Ledoit–Wolf shrinkage rather than the plain sample mean shifted EEGNet's result by +0.8 points on test and −0.2 points on validation, so we kept the simpler estimator. The Riemannian baselines show the largest effect of all: without re-centring, MDM transferred barely above chance (30.2%), and per-session re-centring raised it to 47.1% (+16.9 points, all nine subjects, p = 0.0039).

TABLE XI
EUCLIDEAN ALIGNMENT ABLATIONS (CROSS-SUBJECT ACCURACY, %)

Model, protocol	No EA	EA	Δ	Up	*p*
FBCSP, v1	38.7	44.2	+5.5	7/9	0.074
HCT-Net D, v1 (V3)	41.8	46.9	+5.1	7/9	0.027
MDM, re-centring (RC)	30.2	47.1	+16.9	9/9	0.0039
EEGNet, v2, 1 seed	42.5	51.9	+9.4	9/9	0.0039
EEGNet, v2, shrinkage EA	—	52.6	+10.1	—	—

Δ: gain from alignment. For MDM, alignment is per-session Riemannian re-centring, which is deterministic (no seeds). The shrinkage row uses a Ledoit–Wolf estimate of each session's reference covariance; its Δ is relative to the same no-EA run.

G. Protocol v2, Training Data and the Noise Floor

Before continuing, we measured how much of the v1 numbers was the protocol rather than the models. Under v1, a configuration of HCT-Net with a flattened classifier and global attention (configuration C2 below) scored 56.0% on the validation subjects but 49.1% on the test subjects—a gap of 6.9 points, because the validation subject had also chosen its stopping epoch. Under v2, with stopping moved to the stop-set, the same configuration gave 50.3% on validation and 50.6% on test: the gap closed to 0.3 points, and validation became a faithful guide to test (Table XII).

TABLE XII
PROTOCOL EFFECT AND SEED NOISE (C2 CONFIGURATION)

Run	Val.	Test	Gap
Protocol v1, seed 0	56.0	49.1	6.9
Protocol v2, seed 0	50.3	50.6	−0.3
Protocol v2, seed 1	49.6	49.5	0.1
Protocol v2, seed 2	50.5	50.6	−0.1
Seed spread of the mean (range / SD)	0.9 / 0.46	1.1 / 0.62	
Seed spread of one fold (median / max range)	4.7 / 8.2	3.8 / 4.7	

Gap: validation minus test. The seed controls the initial weights, the random augmentation and the stop-set split.

Repeating the same configuration with three seeds then gave the noise floor. Across seeds the nine-fold validation mean moved by 0.9 points (standard deviation 0.46) and the test mean by 1.1 points, while an individual fold moved by 4.7 points in the median case and by as much as 8.2. Two means from single seeds therefore differ with a standard deviation of roughly 0.65 points, so a single-seed improvement smaller than about 1.3 points lies within seed noise, and differences of less than about five points on a single fold carry no information by themselves. With three seeds per configuration the spread of such a difference falls to about 0.38 points, and the 0.75-point acceptance threshold equals two of these standard deviations. Fig. 12 illustrates the per-fold spread: with nothing changed except the seed, a single validation subject can shift by more than eight points.

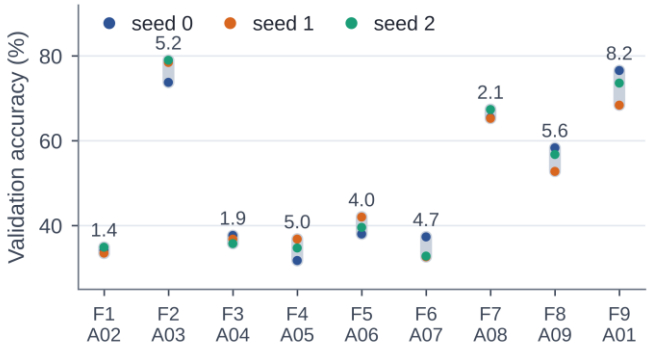


Fig. 12. Seed noise under protocol v2 for the C2 configuration: validation accuracy of each fold for three seeds, which differ only in the initial weights, the random augmentation and the stop-set split. Numbers give the range over seeds. Fk denotes fold k, labelled with its validation subject.

That first estimate came from a single configuration without adaptation, and it proved optimistic. Repeating the calculation for all nine three-seed configurations of protocol v2 (Table XIII) gives a range of the nine-fold validation mean over seeds of 0.40 to 2.68 points and a pooled seed standard deviation of 0.89 points, about twice the first estimate. Two single-seed means therefore differ with a standard deviation of about 1.25 points, and two three-seed means with about 0.72 points, so the 0.75-point acceptance threshold corresponds to roughly one standard deviation of a three-seed difference rather than two. Decisions that cleared or missed the threshold by a wide margin are unaffected. The choice of configuration A, which led the other sizes by 1.0–1.2 points on validation, is supported only moderately by this criterion, and we test it in another way in Section VI-J.

TABLE XIII
SEED NOISE OF EVERY THREE-SEED CONFIGURATION (PROTOCOL V2)

Configuration	Params	Val. range	Val. SD	Test range	Test SD
HCT-Net A	3,924	1.85	1.05	0.95	0.48
HCT-Net B	6,148	1.04	0.60	2.41	1.34
HCT-Net C	12,452	1.81	0.94	1.33	0.72
HCT-Net D	20,996	1.72	0.94	0.15	0.08
C1 (flattened head)	22,276	2.18	1.20	2.62	1.40
C2 (+ global attention)	24,836	1.01	0.55	1.06	0.54
C2 + BN in encoder	24,836	0.40	0.21	1.37	0.69
EEGNet	3,188	2.68	1.36	1.37	0.79
C2, no adaptation (first estimate)	24,836	0.87	0.46	1.08	0.62

Range and sample SD, over three seeds, of the nine-fold mean accuracy (%). All rows except the last use batch-normalisation adaptation. Pooled validation SD: 0.89 points.

We also measured what the separate validation subject costs. With a fixed 64-epoch budget and no stopping, EEGNet trained on seven subjects reached 52.5% on test, and trained on eight subjects (the validation subject added to training, as in Zhao *et al.*'s protocol) 54.3%: +1.8 points from one more subject (single seed, p = 0.20). This is part of the reason our cross-subject numbers sit below published ones (Section VI-P), and it is the price of having an untouched selection subject.

H. Single-Run Improvements Re-Measured

Under protocol v1 we had refined configuration D step by step, retaining any change that increased single-seed validation accuracy (Table XIV, upper part). Two changes were retained: a flattened classifier that sees every time step (C1) and global attention (C2). C2 was not chosen blind: we tried it partly because the component study, whose figures are test-fold accuracies, had shown global attention ahead of windowed attention (49.8% against 46.9%); it was then kept on validation alone, with a gain of 1.0 point over C1. Batch-normalisation re-estimation (C3) was turned down on validation (55.6% versus 56.0%) even though it had the best test score (53.2%), and averaging the softmax outputs of three seeds (C5) produced the final v1 model, with 51.9% on test and 74,508 parameters across the three networks. Both retained architectural changes were single-seed gains of 1.0 to 1.2 points, inside the noise band we measured later.

TABLE XIV
FIRST REFINEMENT ROUND UNDER V1 AND ITS RE-MEASUREMENT UNDER V2

Configuration	Params	Val.	Test	Decision
v1, one seed				
D (initial design)	20,996	53.8	47.1	reference
C1 flatten head	22,276	55.0	48.5	kept
C2 + global attention	24,836	56.0	49.1	kept
C3 + BN adaptation	24,836	55.6	53.2	rejected
C5 3-seed ensemble	74,508	58.6	51.9	final v1
v2, three seeds, BN adaptation on				
D	20,996	53.3	52.7	reference
C1	22,276	53.2	52.5	dropped
C2	24,836	53.6	54.4	dropped
C2 + BN in encoder	24,836	49.3	48.6	rejected

Decisions use validation only. Under v2, C1 vs D −0.1 (better in 4/9, $p = 1.0$) and C2 vs C1 +0.4 (5/9, $p = 0.66$): neither clears the 0.75-point threshold.

When re-measured under v2 with three seeds and batch-normalisation adaptation enabled, neither change held up. On validation C1 scored 0.1 points below configuration D and C2 0.4 points above C1, both well within the threshold, so both changes were dropped and the model returned to its plain form. The cause was an interaction with adaptation, which helped the plain model more than either modified version: on validation it added +5.0 points to configuration D, +4.7 to C1, +3.5 to C2 and +2.4 to C2 with batch normalisation inside the encoder (EEGNet: +2.0). Each change looked useful in isolation but did not combine additively with adaptation—precisely the situation the one-change-per-step rule is meant to expose—and this came to light only because we measured the changes again instead of carrying them forward. For completeness, C2 had the highest test accuracy of the four configurations (54.4%), but the decision was taken on validation, where its lead was 0.3 points. Fig. 13 compares the two rounds: under v1 each step seemed to raise validation accuracy, whereas under v2 the same steps left it unchanged.

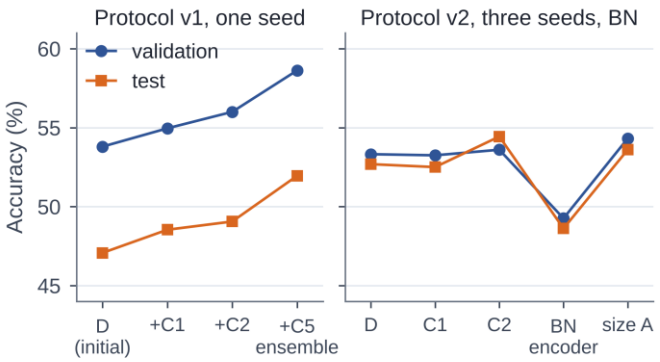


Fig. 13. Validation and test accuracy along the two rounds of model changes. Left: protocol v1, one seed, each change kept on single-run validation gains; the validation–test gap is 6–7 points. Right: protocol v2, three seeds, BN adaptation on; validation and test agree, C1 and C2 bring no gain, batch normalisation in the encoder hurts, and the small configuration A is best.

Swapping the encoder's layer normalisation for batch normalisation, so that test-time adaptation could also act on the attention path, was our own idea and the only architectural change in the study not taken from earlier work. It clearly failed: after adaptation its validation accuracy was 4.3 points lower than that of C2, lower on every one of the nine subjects ($p = 0.0039$), and it gained less from adaptation (+2.4 against +3.5). Early stopping also triggered much earlier (mean best epoch 36 against 110), which fits the explanation that statistics computed over 11-step windows make a noisy normaliser at this batch size.

I. Label-Free Test-Time Adaptation

Table XV compares four adaptation variants, each run with three seeds, for the C2 configuration and for EEGNet. Re-estimating the batch-normalisation statistics (*bn*) increased HCT-Net's validation accuracy by 3.5 points and its test accuracy by 3.7 points, with every one of the nine test subjects improving (Holm-corrected $p = 0.016$). Full TENT added a further 0.3 points on validation, less than the threshold, while requiring gradient updates on the new user, so we kept the simpler variant. Pseudo-labelling gave HCT-Net about one point and EEGNet nothing. For EEGNet, *bn* added 2.0 points on validation and 2.1 on test.

TABLE XV
TEST-TIME ADAPTATION VARIANTS (THREE-SEED MEANS, PROTOCOL V2)

Model, variant	Val.	Δ val	Test	Δ test	Up
HCT-Net C2, none	50.1	—	50.7	—	—
bn (kept)	53.6	+3.5	54.4	+3.7	9/9
TENT	53.9	+3.9	54.1	+3.4	7/9
pseudo-label 0.8	51.0	+1.0	52.3	+1.5	8/9
pseudo-label 0.9	51.2	+1.2	52.2	+1.4	7/9
EEGNet, none	53.0	—	52.2	—	—
bn	55.0	+2.0	54.3	+2.1	6/9
TENT	55.1	+2.1	54.4	+2.2	8/9
pseudo-label 0.8	53.4	+0.3	52.2	+0.0	4/9
pseudo-label 0.9	53.0	−0.1	52.2	0.0	6/9
HCT-Net A, none	49.9	—	49.3	—	—
bn (final)	54.3	+4.4	53.6	+4.3	9/9

Up: test subjects whose three-seed mean improves. Holm-corrected Wilcoxon p (test vs none, four variants): HCT-Net C2 bn 0.016, TENT 0.12; EEGNet bn 0.16, TENT 0.047. HCT-Net A: $p = 0.0039$ (all nine improve).

With the final configuration A, adaptation lifted validation accuracy from 49.9% to 54.3% and test accuracy from 49.3% to 53.6%, and once more every subject improved ($p = 0.0039$). The benefit was greatest where it was most needed (Fig. 14): the four subjects with the lowest accuracy before adaptation gained 4.9 points on average, while the four with the highest gained 1.8 points, with the largest single gain, 11.7 points, on A07. Because the subjects were ranked by their unadapted accuracy, part of this pattern could be regression to the mean. Ranking the subjects on one seed and measuring the gain on the other two gives the same picture (5.3 against 1.8 points), and unadapted accuracy and gain are negatively correlated across subjects ($r = -0.49$), so the pattern is not an artefact of ranking on noisy scores. For EEGNet the gains were smaller and less uniform (+2.1 on average, with three subjects slightly worse). The step takes about 0.2 s per subject on the GPU and uses no label from the new user.

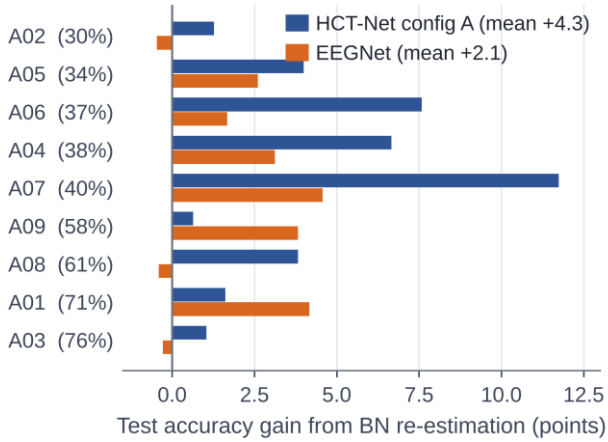


Fig. 14. Per-subject test-accuracy gain from batch-normalisation re-estimation for HCT-Net configuration A and EEGNet (three-seed means, protocol v2). Subjects are ordered by HCT-Net's unadapted accuracy, given in parentheses; the weakest subjects gain most.

How much data each variant actually adapts on helps explain the ranking. Pseudo-labelling selects trials with the unadapted model's own confidence: at a threshold of 0.8 it retained on average 48.6% of a test subject's 576 trials for HCT-Net but only 23.4% for EEGNet, and at 0.9 it retained 28.8% and about 10%; in 3 EEGNet runs fewer than 16 trials qualified and the model fell back to its unadapted predictions. Self-training thus sees a confidence-selected minority of the new user's data, whereas batch-normalisation re-estimation uses all of it and needs no threshold. HCT-Net was confident far more often than EEGNet although it was slightly less accurate before adaptation (50.7% against 52.2%), a pattern that points to less well calibrated output probabilities; the logs do not record the accuracy of the retained pseudo-labels, so this could not be checked directly.

One disclosure is needed. We ran the adaptation experiment earlier than planned because the v1 test result for C3 had looked promising. The decision to keep or reject a variant was nevertheless based on validation alone, and the variants and their settings were fixed before the run.

J. Size Sweep

By this point the evidence pointed one way: 81% of configuration D's parameters were in the encoder, the component study had found attention unhelpful, EEGNet led with only 3,188 parameters, and each addition to the plain model had reduced what adaptation could contribute. We therefore trained the four sizes listed in Table V, with three seeds each, under protocol v2 with adaptation enabled, and also measured the within-subject accuracy of each size with the same recipe (Table XVI, Fig. 15). Configuration A (one encoder layer, width 16, 3,924 parameters) beat each of the other sizes on validation by 1.0 to 1.2 points and was ahead on seven or eight of the nine subjects, so it was selected outright; it is also the smallest. The other three configurations differed by no more than 0.2 points: adding a second encoder layer brought no gain (B against A), and neither did doubling the width (C against A, D against B). How much adaptation helped did not change systematically with size (+3.8 to +5.0 points on validation).

TABLE XVI
SIZE SWEEP (PROTOCOL V2, THREE SEEDS, BN ADAPTATION)

Config	Params	Val.^{0}	Val.	Test	κ	Within	Gap
A	3,924	49.9	54.3	53.6	0.381	72.1	18.5
B	6,148	48.4	53.2	53.5	0.380	71.9	18.4
C	12,452	49.4	53.1	52.5	0.367	73.4	20.9
D	20,996	48.4	53.3	52.7	0.369	73.2	20.6

Val.^{0}: validation without adaptation. Within: session T → E accuracy with the same recipe. Gap: within minus cross-subject test. Pairwise on validation: A vs D +1.00 (better in 7/9, $p = 0.16$), A vs B +1.14 (8/9, $p = 0.098$), A vs C +1.18 (7/9, $p = 0.074$), B vs D −0.14, C vs D −0.18.

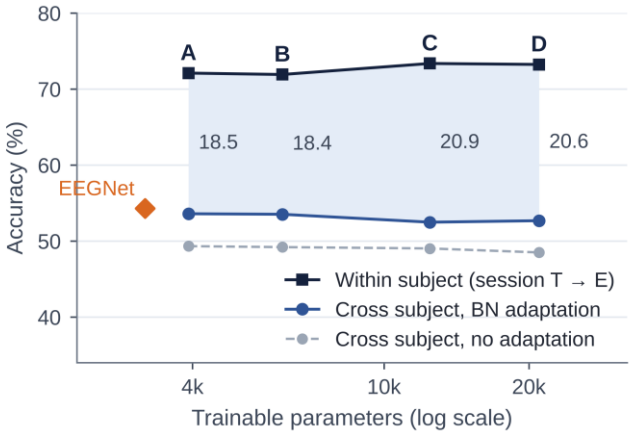


Fig. 15. Within-subject and cross-subject accuracy of the four HCT-Net sizes (three seeds each). Numbers give the gap between a familiar and an unseen user. Reducing the model from 20,996 to 3,924 parameters narrows the gap by about two points; about 18 points remain at every size. The orange marker is EEGNet under the same protocol.

Two checks qualify this choice. First, the lead of 1.0–1.2 points is only about one and a half standard deviations of the seed noise measured across configurations (Section VI-G). Second, the nine validation subjects are the same people as the nine test subjects, so a selection made on all of them could favour configurations that happen to suit these particular people. To test this we repeated the choice nine times, each time leaving out the fold whose validation subject is the subject being tested, and scored the chosen configuration on that subject. Configuration A was chosen in all nine repetitions, both among the four sizes and among all seven HCT-Net configurations of protocol v2, so its test accuracy of 53.6% is unchanged under this nested selection. With EEGNet added to

the candidates, the same procedure picks EEGNet every time (54.3%).

The size sweep also shows how much of the gap between familiar and unseen users is due to capacity. Within-subject accuracy hardly changed across the four sizes (71.9–73.4%), so any change in the gap came from the cross-subject side: with adaptation the gap was 20.6 points at 20,996 parameters and 18.5 points at 3,924 (24.7 and 22.8 without adaptation). Model size therefore explains about two points of the gap. The remaining 18 points persist at every size we tried and cannot be removed by making the model smaller still. As a consistency check, the within-subject mean of configuration D, 73.2%, is identical to that of the earlier within-subject run of the same model, confirming that the within-subject protocol was applied in the same way both times. One caveat applies to every gap in this paper: within-subject accuracy is scored on session E only, whereas cross-subject accuracy is scored on both sessions of the test subject. The per-trial predictions needed to restrict the cross-subject score to session E were not logged, so part of each gap may reflect a difference between sessions rather than between subjects.

Per subject, the gap for configuration A ranges from 2.5 points (A06) to 35.3 points (A07) (Fig. 16). It is not simply larger for the subjects who are hard to decode: A07 is decoded well within subject (86.9%) but only moderately across subjects (51.6%), whereas A06 is decoded poorly in both settings. A small gap therefore does not indicate a good calibration-free user, only that the user's own calibration data would not help much.

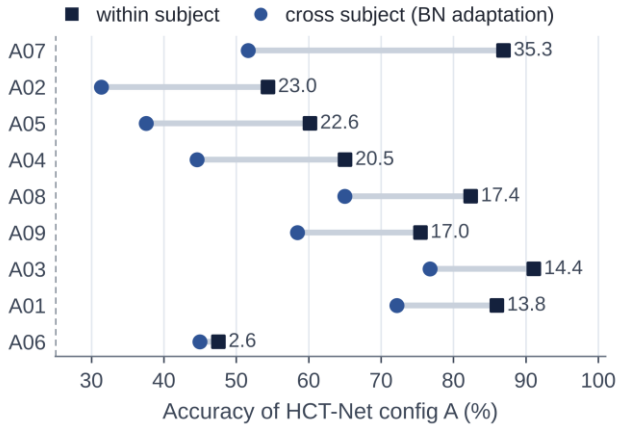


Fig. 16. Within-subject and cross-subject accuracy of HCT-Net configuration A for each participant (three-seed means), ordered by the size of the gap (numbers, in points). The dashed line marks chance.

K. Final Comparison

Table XVII brings together the final models, all evaluated under protocol v2, each with three seeds and with adaptation for every model. HCT-Net configuration A achieved 53.6% ± 15.6 test accuracy (κ = 0.381) and 54.3% on validation, and EEGNet 54.3% ± 17.0 (κ = 0.391) and 55.0% on validation. The 0.7-point validation advantage of EEGNet is smaller than our 0.75-point threshold, and paired tests show no difference on either split (test: −0.7 points, HCT-Net ahead on 3 of the 9 subjects, p = 0.50; validation: −0.7 points, p = 0.50). We therefore treat the outcome as a draw. Subject by subject (Fig. 17) the two models follow each other closely; the largest gaps are on A07, where HCT-Net leads by 9.1 points, and on A09, where EEGNet leads by 8.5 points.

TABLE XVII
FINAL CROSS-SUBJECT RESULTS (PROTOCOL V2, THREE SEEDS, BN ADAPTATION) AND RIEMANNIAN BASELINES

Model	Params	Val.	Test ± SD	κ
HCT-Net A (ours)	3,924	54.3	53.6 ± 15.6	0.381
EEGNet	3,188	55.0	54.3 ± 17.0	0.391
HCT-Net D (initial design)	20,996	53.3	52.7 ± 15.3	0.369
Riemannian, deterministic, no selection:				
TS-LR + re-centring	—	—	50.5 ± 15.4	0.340
MDM + re-centring	—	—	47.1 ± 14.6	0.295

Chance accuracy is 25%. SD: sample standard deviation across the nine test subjects. The Riemannian baselines use the same folds and seven training subjects but no validation subject. The larger deep baselines were run only under protocol v1 (Table VIII).

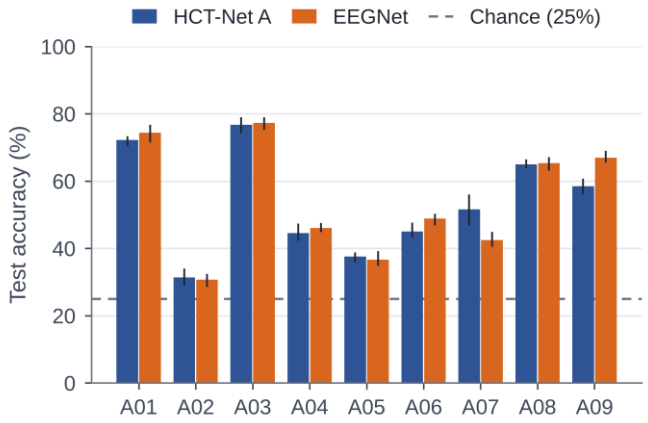


Fig. 17. Per-subject test accuracy of HCT-Net configuration A and EEGNet under protocol v2 with adaptation (three-seed means; vertical lines show the range over seeds). The two models are statistically level (p = 0.50).

The two Riemannian baselines with re-centring, which use no seeds and no selection subject, reached 50.5% (TS-LR) and 47.1% (MDM). Configuration A was 3.1 points ahead of TS-LR (better on 6 of 9 subjects, p = 0.098), and EEGNet 4.4 points ahead (p = 0.039, uncorrected), so the compact networks are ahead of a strong classical reference but not by a margin nine subjects can establish firmly. The larger deep baselines (ATCNet, EEG Conformer and CTNet) were evaluated only under protocol v1 (Table VIII), where they reached 50.2–52.4% without adaptation. They were not re-run with the corrected protocol and adaptation, so how they would fare under the conditions of Table XVII is unknown. The final HCT-Net is 5.4 times smaller than configuration D and 178 times smaller than EEG Conformer.

L. Effect Sizes, Uncertainty and Power

Nine test subjects make every comparison in this paper a small-sample comparison, and with small samples a non-significant difference is weak evidence of no difference [50]. Table XVIII therefore reports, for the comparisons behind our main conclusions, the mean of the per-subject differences with a 95% confidence interval (t distribution, eight degrees of freedom), the number of subjects with a positive difference, and the minimum detectable effect (MDE): the smallest mean difference that the exact signed-rank test would have detected with 80% power, given the observed spread of the differences.

With nine pairs, the two-sided test at α = 0.05 rejects only when the smaller signed-rank sum is at most 5 (exact size 0.039), and the smallest attainable *p*-value is 2/512 = 0.0039, reached only when all nine subjects move in the same direction. Simulating normally distributed differences (200,000 draws per effect size), the test reaches 80% power when the true mean difference is 1.15 standard deviations of the differences; at α = 0.01, the level that the smallest of five Holm-corrected *p*-values must reach (Table IX), it needs 1.61 (Fig. 18). Holm correction over the eight comparisons of Table XVIII is stricter still: the smallest *p*-value must fall below 0.05/8 = 0.0063, which only a unanimous nine-subject result can reach.

TABLE XVIII
PAIRED DIFFERENCES BEHIND THE MAIN CONCLUSIONS

Comparison (first − second)	Mean	95% CI	*p*	Holm	Up	MDE
(1) A − EEGNet, v2	−0.7	−4.3, 2.9	0.50	0.85	3/9	5.4
(2) A − D, v2	+0.9	−0.9, 2.8	0.43	0.85	5/9	2.8
(3) A: BN adapt. − none	+4.3	1.4, 7.1	0.0039	0.031	9/9	4.3
(4) EEGNet: BN adapt. − none	+2.1	0.5, 3.7	0.055	0.22	6/9	2.3
(5) EEGNet: EA − no EA	+9.4	4.7, 14.0	0.0039	0.031	9/9	7.0
(6) FBCSP: EA − no EA, v1	+5.5	−0.6, 11.5	0.074	0.22	7/9	9.0
(7) HCT-Net D − EEGNet, v1	−4.3	−7.8, −0.7	0.023	0.12	1/9	5.3
(8) A: within − cross subject	+18.5	11.7, 25.3	0.0039	0.031	9/9	10.1

Accuracy differences in points, per test subject: three-seed means for rows 1–4 and 8 (with BN adaptation except where adaptation is the comparison), one seed for rows 5–7; within-subject accuracy is on session E; row 7 is configuration D under protocol v1. 95% CI: *t* interval. *p*: two-sided Wilcoxon signed-rank test (Section V-E), uncorrected; Holm: corrected across the eight comparisons. Up: subjects with a positive difference. MDE: 1.15 × the SD of the differences (80% power, α = 0.05). Row 4: the interval excludes zero but the rank test does not reach 0.05, because three subjects lost accuracy.

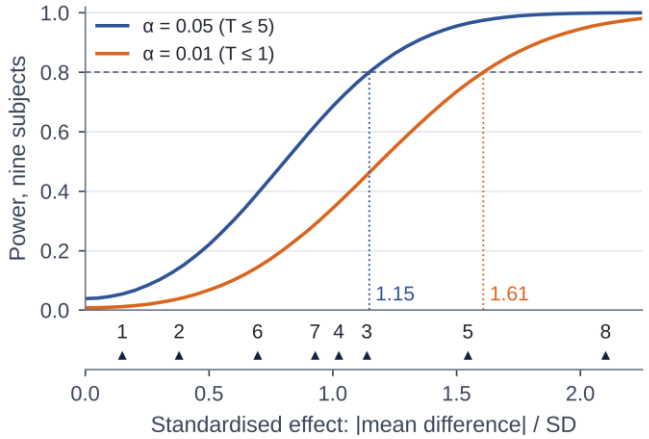


Fig. 18. Power of the exact two-sided Wilcoxon signed-rank test with nine subjects as a function of the standardised paired effect (mean difference divided by the SD of the differences), for normally distributed differences (200,000 simulations per point). Dotted lines: 80% power is reached at 1.15 SD for α = 0.05 and at 1.61 SD for α = 0.01. Lower strip: the observed standardised effects of the comparisons in Table XVIII, numbered by row. They show where each comparison lies relative to what the design could detect, not the power the test had.

Three conclusions are well supported. Euclidean alignment for EEGNet (+9.4 points, 95% CI 4.7 to 14.0), batch-normalisation adaptation for configuration A (+4.3, 1.4 to 7.1) and the gap between within- and cross-subject accuracy (18.5, 11.7 to 25.3) are large relative to their spread, every subject moved in the same direction, and all three remain significant after Holm correction across the eight comparisons (corrected p = 0.031 each). The null results need more care. The tie between configuration A and EEGNet (−0.7 points, −4.3 to 2.9) does not show that the two are equivalent. It bounds configuration A's advantage at about three points and its deficit at about four, and the study had 80% power only for differences larger than 5.4 points.

The comparison of configurations A and D is more informative than its *p*-value suggests. Because the two configurations' per-subject results move together (SD of the differences 2.4 points), the interval for A − D is narrow (−0.9 to 2.8): on the test subjects, the configuration with 5.4 times as many parameters is not better than configuration A by more than about one point. This is the strongest quantitative form of our central claim. The v1 deficit of configuration D against EEGNet (−4.3, −7.8 to −0.7) is about the size of its MDE (5.3 points), which is why it was significant before correction but not after. At the other extreme, alignment for FBCSP (+5.5, −0.6 to 11.5) is an effect that may well be real but that nine subjects with this much spread (SD 7.8 points) cannot establish: detecting it reliably would have required a mean effect of about 9 points.

M. Chance Levels and Information Transfer

Two further quantities put the accuracies in context. First, an accuracy above 25% is not automatically above chance. With 576 test trials per subject, a one-sided binomial test at α = 0.05 requires 28.1% correct, and 29.9% after Bonferroni correction over the nine subjects [51], [52]; for the 288 trials of a within-subject test session the thresholds are 29.5% and 31.9%. Every model in Table XIX exceeded the uncorrected threshold for every subject. With the correction, every protocol v2 model still did so, although A02 came close (30.1–34.0% across the four v2 cross-subject rows). These statements concern three-seed means; in single runs, A02 fell below the corrected threshold once for configuration A (29.2%) and once for EEGNet (28.8%). Under protocol v1, A02 fell below the corrected threshold for FBCSP, EEGNet and HCT-Net D (28.3–29.2%), as did A05 for FBCSP (29.3%); for these subject–model pairs, the output cannot be distinguished from guessing once all nine subjects are tested.

Second, accuracy alone understates how far a four-class decoder at 53.6% is from a usable interface. Wolpaw's information transfer rate [1] gives the bits conveyed per selection by an M-class decoder that is correct with probability P_c:

$$B = log_2 M + P_c log_2 P_c + (1 - P_c) log_2 \left(\frac{1-P_c}{M-1}\right) \qquad (9)$$

It assumes equiprobable classes and errors spread evenly over the wrong classes. Our errors are not uniform (Section VI-N), so the values in Table XIX are approximate. Averaged over subjects, configuration A transfers 0.33 bits per trial across subjects and 0.79 within subject, 2.4 times as much. At one selection per 6 s—fixation onset to the end of the analysis window, ignoring the inter-trial break—these correspond to 3.3 and 7.9 bits/min. Because B is convex in P_c, the subject mean (0.33) is higher than the rate at the mean accuracy (0.27), and the spread is wide: 0.01 bits per trial for A02 against 0.85 for A03. Most of the information a calibration-free decoder delivers therefore goes to a few users, which is the practical meaning of the large between-subject SD.

TABLE XIX
CHANCE-LEVEL CHECKS AND INFORMATION TRANSFER RATE

Model, setting	Acc.	Above	Bits	Bits/min
Cross subject, protocol v2:				
HCT-Net A, BN adaptation	53.6	9/9	0.33	3.3
HCT-Net A, no adaptation	49.3	9/9	0.27	2.7
EEGNet, BN adaptation	54.3	9/9	0.36	3.6
HCT-Net D, BN adaptation	52.7	9/9	0.31	3.1
Cross subject, protocol v1:				
FBCSP + LDA	44.2	7/9	0.17	1.7
EEGNet	51.2	8/9	0.30	3.0
ATCNet	52.4	9/9	0.33	3.3
EEG Conformer	52.1	9/9	0.30	3.0
CTNet	50.2	9/9	0.26	2.6
HCT-Net D	46.9	8/9	0.24	2.4
Within subject, session E:				
HCT-Net A (v2 recipe)	72.1	9/9	0.79	7.9
EEGNet (v1)	65.9	9/9	0.61	6.1
EEG Conformer (v1)	77.3	9/9	0.93	9.3

Acc.: mean accuracy (%). Above: subjects significantly above chance (one-sided binomial test, Bonferroni-corrected over nine subjects: 29.9% for 576 test trials, 31.9% for 288). Bits: Wolpaw information transfer rate per trial, averaged over subjects. Bits/min: at one selection per 6 s.

N. Error Analysis

1) Classes: Fig. 19 shows pooled confusion matrices for HCT-Net D and EEGNet under protocol v1, the last protocol for which per-trial predictions were logged for these models. Feet was the hardest class for every model: its recall ranged from 28.8% to 41.5% across the nine v1 configurations, and its most frequent confusion was with tongue (30% of feet trials for HCT-Net, 24% for EEGNet). Both are classes without a strongly lateralised signature (Fig. 5). Hands behaved differently for the classical and the deep models. For FBCSP, 25.9% of hand trials were assigned to the other hand, slightly more than the 23.6% of feet and tongue trials assigned to each other, so the hemispheric lateralisation that makes hands easy within a subject did not transfer. The learned spatial filters of the deep models recovered most of that advantage: EEGNet assigned 14.8% of hand trials to the other hand against 26.3% for feet and tongue. No model collapsed onto one class; predictions were spread almost uniformly over the four classes. The confusion matrices of all six v1 models are given in Fig. 22 and their per-class precision, recall and F1 in Table XXIX (Appendix A): feet has the lowest F1 of the four classes for every model (32.6–40.7%).

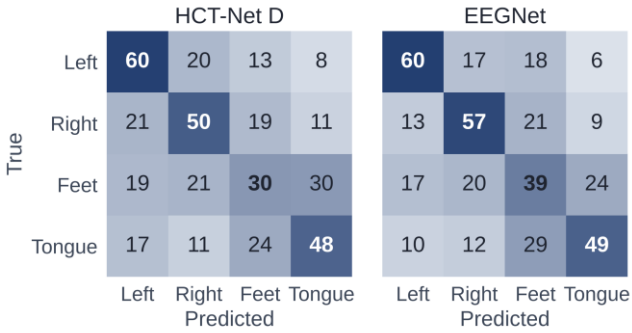


Fig. 19. Pooled cross-subject confusion matrices (protocol v1; percentage of each true class, 1,296 trials per class) for HCT-Net D and EEGNet. Feet is confused with tongue more than any other pair.

2) Subjects: The hardest subjects were the same for every model and every protocol: A02 and A05, followed by A04 and A06, whereas A03 and A01 were the easiest (Fig. 20). In no reported configuration (single runs under v1, three-seed means under v2) did A02 exceed 35% or A05 exceed 43%. To understand why, we related each subject's accuracy to the session-to-session covariance distance of Section III-C. The relationship is strong (Fig. 21): subjects whose own two recording days differ more are harder to decode cross-subject (configuration A: $r = -0.89$, 95% CI −0.98 to −0.54, $p = 0.001$; EEGNet: $r = -0.89$, −0.98 to −0.54, $p = 0.001$) and within subject ($r = -0.75$, −0.94 to −0.17, $p = 0.020$). A02 and A05 have the largest session distances (3.38 and 3.01) and the lowest accuracies; A01 has the smallest distance (1.03) and is among the best. Because Euclidean alignment equalises second-order statistics by construction, the variability that survives it—and that predicts failure—must lie in structure that the mean spatial covariance does not capture. This analysis was not planned in the design and should be read as exploratory.

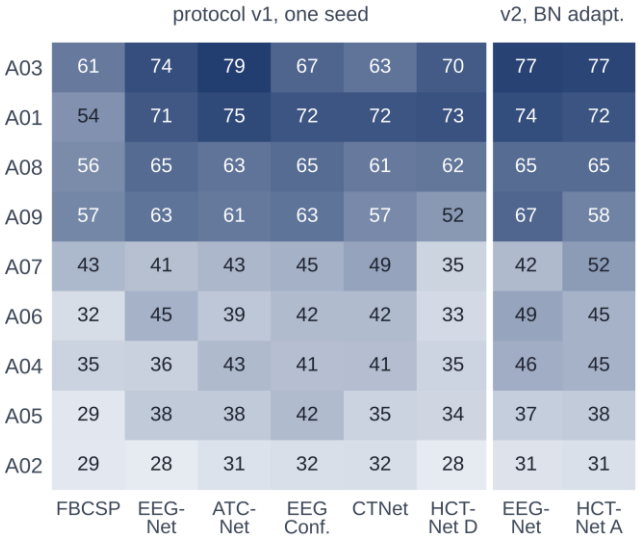


Fig. 20. Cross-subject test accuracy (%) of every subject under eight model–protocol combinations, subjects ordered by mean accuracy. The ranking of subjects is almost the same for every decoder, classical or deep, large or small.

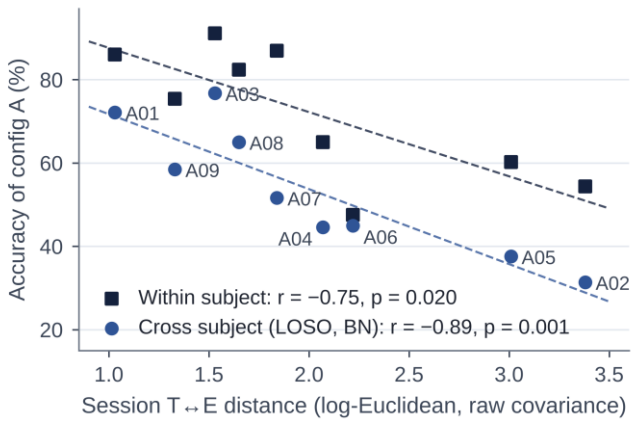


Fig. 21. Per-subject accuracy of HCT-Net configuration A against the log-Euclidean distance between the subject's two sessions' mean covariances (before alignment). Dashed lines: least-squares fits. Subjects whose recordings change most between days are hardest to decode, both within and across subjects (exploratory analysis).

A09 deserves a separate note. It is the subject on which EEGNet beats HCT-Net by the most (67.0% against 58.5%), and it is also the subject whose training data lost 25% of trials—half of its tongue trials—to the amplitude threshold when it served as a training subject. Its own test trials were never removed.

O. Computational Cost

Table XX reports parameter counts and measured training times. Under protocol v1 a LOSO fold took on average 213 s for EEGNet, 216 s for HCT-Net D, 325 s for CTNet, 372 s for EEG Conformer and 575 s for ATCNet on the RTX 4050, including early stopping; FBCSP took 52 s on the CPU. Under protocol v2, which stops on the stop-set and therefore tends to train longer, configuration A took 247 s per fold (best epoch 162 on average) and EEGNet 108 s (best epoch 55). Test-time adaptation added about 0.2 s per subject. The per-fold logs add up to about 36 hours of training for every experiment reported here, all on one laptop GPU, which places a study of this kind within reach of a student project.

TABLE XX
MODEL SIZE AND TRAINING COST

Model	Params	v1 s/fold	v2 s/fold
HCT-Net A (ours)	3,924	—	247
EEGNet	3,188	213	108
HCT-Net D	20,996	216	343
ATCNet	113,732	575	—
CTNet	152,364	325	—
EEG Conformer	697,412	372	—
FBCSP + LDA	—	52	—

Mean wall-clock time per LOSO fold on an NVIDIA RTX 4050 (6 GB); FBCSP on the CPU. v2 times are means over three seeds and include training only.

Training length also differed between model families (Table XXI). None of the 216 protocol v2 training runs in the table reached the 500-epoch cap—the latest best checkpoint was at epoch 347—so the epoch budget did not limit any result. EEGNet reached its best stop-set accuracy after a median of 41 epochs, against 96–144 for the HCT-Net configurations, which accounts for most of its shorter training time. Within the HCT-Net family the best epoch showed no trend with size. The exception was the batch-normalised encoder, which peaked after a median of 25 epochs and in one run after 7: the early-stopping signature noted in Section VI-H.

TABLE XXI
BEST EPOCH UNDER PROTOCOL V2 (27 RUNS PER MODEL)

Model	Params	Median	IQR	Range
EEGNet	3,188	41	31–77	18–159
HCT-Net A	3,924	144	113–209	66–347
HCT-Net B	6,148	111	75–156	53–302
HCT-Net C	12,452	134	103–170	37–226
HCT-Net D	20,996	129	102–178	62–243
C1 (flattened head)	22,276	103	66–149	22–240
C2 (+ global attention)	24,836	96	73–145	31–259
C2 + BN in encoder	24,836	25	16–51	7–109

Best epoch: the epoch whose checkpoint had the highest stop-set accuracy; training continued for 50 more epochs before stopping. Nine folds × three seeds per model. IQR: interquartile range. Budget: 500 epochs.

P. Published Figures and Their Protocols

Comparisons with published numbers are meaningful only when the protocols match, and the figures most often quoted for these models frequently come from within-subject evaluations. Table XXII lists them together with the protocol that produced each one. The values commonly cited for EEGNet (68.67%) and ATCNet (81.10%) come from the ATCNet authors' code repository, where a within-subject train–validation–test script produces them, and EEG Conformer's 78.66% is a within-subject session hold-out result. The only like-for-like cross-subject source we found is Zhao *et al.* [8], who report CTNet at 58.64% and their re-implementations of EEGNet and EEG Conformer at 56.85% and 53.41% under leave-one-subject-out evaluation. Parameter counts depend on the input as well: EEGNet has 2,548 parameters for roughly 2.2 s of input and 3,188 for our 3.5 s.

TABLE XXII
FREQUENTLY QUOTED FIGURES AND THE PROTOCOLS BEHIND THEM

Figure	Source	Protocol that produced it	Verdict
CTNet 82.52% / 58.64% (IV-2a)	Zhao *et al.* [8]	within subject / LOSO with 8 training subjects, 2–6 s, 600 epochs	verified
CTNet 88.49% / 76.27% (IV-2b)	Zhao *et al.* [8]	within subject / LOSO	verified
EEGNet+ 56.85%, EEG Conformer+ 53.41%	Zhao *et al.* [8]	LOSO re-implementations, same protocol as CTNet	verified
EEG Conformer 78.66%	authors' repository [11]	session hold-out, within subject	verified
ATCNet 81.10%; EEGNet 68.67%	ATCNet repository	within-subject train–validation–test script	verified, but within subject
ATCNet 85.38%	Altaheri *et al.* [12]	subject-dependent	verified

Against the one verified like-for-like LOSO source, our reproductions sit 1.3 points (EEG Conformer) and 2.6 points (EEGNet, eight training subjects, fixed epochs) below the published re-implementations, and 8.4 points below for CTNet.

Against this like-for-like source, our EEGNet is 2.6 points lower under the most similar protocol and our EEG Conformer 1.3 points lower; only CTNet, 8.4 points lower, is a genuine outlier. Published cross-subject accuracies are therefore far below the within-subject numbers that are often cited next to them, and our own results are in line with them.

VII. DISCUSSION

A. Extra Capacity Did Not Help Here

Our main finding about model size is negative (RQ1), and its limits matter as much as the finding. Under protocol v1, the four strongest deep architectures, whose parameter counts differ by more than two orders of magnitude, reached cross-subject accuracies within 2.2 points of one another; under the corrected protocol v2, HCT-Net configurations from 3,924 to 24,836 parameters were within two points, and reducing the model from 20,996 to 3,924 parameters lost nothing. Where additional capacity did pay off, it paid off within subject: EEG Conformer and HCT-Net D were the two strongest within-subject decoders, yet they transferred no better than EEGNet. Two qualifications prevent a stronger claim. The large baselines were evaluated only under protocol v1, with one seed and without adaptation, and two of them, ATCNet and CTNet, did not reproduce their published within-subject accuracy under our shared training recipe; better-tuned large models might transfer better than ours did. What the data support is therefore narrower: across the models and settings we could run, extra capacity bought no measurable cross-subject accuracy, and the surplus appears to go into patterns peculiar to the training subjects.

What, then, does limit performance (RQ4)? Two observations implicate the data. First, the 18-point difference between familiar and unseen users remained at every model size while within-subject accuracy did not move, so the gap belongs to the task rather than to any particular model. Second, the size of the gap differs greatly between people and is predicted by how much each person's recordings change from one day to the next. A LOSO model learns from seven training subjects, that is, seven draws from the population of users, and extra capacity cannot help it generalise to someone whose signals fall outside the range those seven cover. The remaining levers therefore act on what the model sees rather than on how much it can memorise: more training subjects, better alignment of the input distributions and adaptation to the new user. The +1.8 points that EEGNet gained from a single extra training subject (Section VI-G) is a small but direct measurement of the first of these.

B. What Transfers: Second-Order Statistics, Not Attention

Both components that reliably helped on unseen users are label-free re-normalisations (RQ2, RQ3). Euclidean alignment equalises the average spatial covariance of the input, and re-estimating batch-normalisation statistics equalises the means and variances of the intermediate features. Between them they explain most of the improvement from configuration D to the final model, with alignment contributing 5 to 9 points for the deep models and 17 points for the Riemannian MDM classifier, and adaptation another 2 to 5. The alignment results are in line with the large evaluation by Junqueira *et al.* [25], and the adaptation results with the AdaBN idea [27] and with online test-time adaptation for MI [29]. Adding gradient updates (TENT) gained little beyond re-estimating the statistics, which suggests that most of what adaptation corrects here is a change in the scale and offset of the features rather than a shift of the decision boundary.

Attention, after which HCT-Net is named, brought no benefit in the experiments that tested it. Removing it raised cross-subject accuracy under v1; global attention outperformed the windowed form and so contradicted the locality assumption of our design; and with adaptation enabled, neither variant made a difference. Every modification of the plain model reduced the gain from adaptation, from +5.0 points on validation for the plain model to +3.5 with global attention. HCT-Net also gained more from adaptation than EEGNet did (+3.7 against +2.1 on test, and more on seven of the nine subjects), but this difference was not significant (p = 0.30). Adaptation acts only on the convolutional front end, because layer normalisation in the encoder keeps no running statistics, and our attempt to extend it to the encoder with batch normalisation failed because it degraded the trained model. The decisive experiment, configuration A without its encoder under protocol v2, was not run, so we cannot say whether the attention layer of the final model contributes anything at all. We therefore do not claim that attention cannot help cross-subject decoding; at this scale of data and with these architectures it did not, and gains measured within subject do not establish its value.

C. Within-Subject Accuracy Was Not a Useful Proxy

Across the six models, within-subject accuracy did not predict cross-subject accuracy ($r = 0.08$). With six models the 95% interval for that correlation runs from −0.78 to 0.84, so we cannot rule out a moderate relationship; what we can say is that within-subject rankings gave no reliable guide to cross-subject ones here. Leaderboards that rank MI decoders by within-subject accuracy therefore say little about calibration-free use, the setting that motivates most of them. We also fell into a related trap: relating within-subject accuracy to the drop in cross-subject accuracy produced an impressive and significant correlation ($r = 0.86$) that arises only because the drop is itself computed from within-subject accuracy. When a quantity is related to a difference that contains it, the test should be run on the components directly.

D. Protocol Choices Matter as Much as Architecture

Several conclusions shifted when the protocol changed, by amounts as large as the differences between architectures. Allowing the validation subject to pick the stopping epoch inflated validation accuracy by 6.9 points. Holding out 20% of the within-subject training data cost the weakest participant 29.9 points. Single-seed gains of about one point, enough to keep two architectural changes under v1, disappeared over three seeds, and per-fold seed noise of up to 8 points exceeds most effects reported in the literature. Our own acceptance threshold, calibrated on one configuration, turned out to be about half as strict as intended once seed noise was measured on all of them. None of this is particular to our models. From this experience we suggest the following practice for cross-subject MI studies: (i) use separate data for early stopping and for model selection, and never select on the test subject; (ii) measure the seed-noise floor on several configurations first and derive acceptance thresholds from it; (iii) introduce changes individually and measure accepted changes again whenever the pipeline changes; (iv) check that the final selection survives leaving each test subject out of it; (v) report results per subject, with paired tests across subjects and both corrected and uncorrected p-values [33]; and (vi) cite published figures together with the protocol behind them.

E. Subjects, Sessions and the Residual Gap

How hard a subject was to decode hardly depended on the decoder: the same participants were the most difficult for a classical and four deep models, within and across subjects and under both protocols. The exploratory link between difficulty and the covariance shift between a participant's two sessions ($r = -0.89$, 95% CI −0.98 to −0.54, nine subjects) ties the cross-subject problem to the cross-session one, since someone whose own signals change from day to day is unlikely to resemble other people consistently. The analysis was not planned and rests on nine points, so it is a hypothesis rather than a finding. Euclidean alignment sets the mean covariance of every session to the identity, so the variability that is left, and that predicts failure, has to lie in structure beyond the mean covariance: for example the spectral shape of the mu rhythm, the spatial layout of its desynchronisation or the imagery strategy the participant uses. This agrees with studies of performance variation and BCI inefficiency [5], [7], and suggests that alignment matching higher-order statistics, or a quick check of a new user's session stability, may achieve more than further architectural changes.

F. Implications for Calibration-Free BCIs

Four-class accuracy of 53.6% ($\kappa = 0.381$) without any calibration is clearly above chance, yet below what most communication applications require, and the mean conceals a wide spread: three participants were decoded at 65–77% without a single one of their labels, while two remained between 31% and 38%. Measured as information, the contrast is even sharper. Configuration A conveys, averaged over subjects, about 0.33 bits per trial to an unseen user and 0.79 to a calibrated one (the rate of the mean accuracy is 0.27 bits, the bit rate being convex in accuracy), and the range across users (0.01 to 0.85 bits) is far wider than any difference between the models we tested (Section VI-M). A practical system could thus run without calibration for the users for whom this works and ask the others for a short calibration session; the label-free adaptation takes a fraction of a second and could use an unlabelled warm-up block. With fewer than four thousand parameters the final model is also small enough for low-power embedded hardware, although we did not measure its inference cost on such devices.

G. Relation to Prior Cross-Subject Results

Our cross-subject accuracies are below the subject-independent results that Zhao *et al.* report for the same architectures [8], and the protocol explains most of the difference. Their leave-one-subject-out evaluation trains on all eight other subjects for a fixed 600 epochs with a 4 s window and keeps no subject aside for selection, whereas ours reserves a validation subject, stops early and uses 3.5 s of signal. Under the most similar protocol we could run, with eight training subjects and a fixed number of epochs, EEGNet reached 54.3% compared with the 56.85% they give for their EEGNet variant, and our EEG Conformer under protocol v1 was 1.3 points below their value. CTNet, 8.4 points lower, is the only large discrepancy, and its within-subject reproduction also broke down on weak subjects. Our alignment results point in the same direction as the large evaluation of Junqueira *et al.* [25], who found consistent gains from Euclidean alignment in deep cross-subject decoding, and our adaptation results agree with test-time adaptation studies for MI [29], except that we adapt offline on a block of unlabelled trials instead of trial by trial.

H. Design Assumptions Revisited

Since the design was fixed before any experiment, each of its assumptions can be compared with the measurements (Table XXIII). Of the nine assumptions, two were confirmed, one only after the protocol had been corrected, two were harmless but uninformative, and four were contradicted. We see this as the most valuable output of a pre-specified study, because every failed assumption is a finding that a design adjusted after the fact would have concealed.

TABLE XXIII
DESIGN ASSUMPTIONS AND WHAT THE MEASUREMENTS SHOWED

Assumption in the design	Evidence	Verdict
Fewer than 50,000 parameters limit subject-specific fitting	accuracy flat from 3,188 to 697,412 parameters; HCT-Net best at 3,924	right about transfer, irrelevant to accuracy
Windowed attention suits short trials	global attention +2.9 over windowed (v1); no difference after adaptation (v2)	not supported
Attention improves cross-subject decoding	removing attention +1.9 (v1, 8 of 9 subjects)	not supported
Two encoder layers are sufficient	depth 1, 2, 6 within 1.7 points; one layer best in size sweep	harmless, one layer suffices
S&R augmentation is necessary on a small dataset	removing it +1.2, $p = 0.91$	no measurable effect
Euclidean alignment removes subject-specific mixing	gain +5.1 (HCT-Net D), +5.5 (FBCSP), +9.4 (EEGNet), +17.1 (MDM); subject recovery 46.8% → 22.9%	supported, second order only
A held-out validation subject guides selection	validation–test gap 6.9 points (v1), 0.3 points (v2)	supported after the protocol fix
Within-subject accuracy indicates decoder quality	within vs cross $r = 0.08$ across six models	not supported
Batch normalisation in the encoder extends adaptation (added during development)	−4.3 on validation, worse in 9 of 9 subjects	rejected

I. Ethical Considerations

The study relied on a public, de-identified dataset used under its published licence, and no new data were recorded. Calibration-free decoding is motivated by assistive applications, in which inflated accuracy claims can create false hopes for patients and carers. We therefore show results for every subject, so that the users for whom the system would fail remain visible instead of being averaged out; we call ties ties; and we retain the negative results.

VIII. LIMITATIONS AND THREATS TO VALIDITY

Sample size and dataset. Every result comes from a single dataset of nine participants. With nine paired observations the exact signed-rank test reaches 80% power only for mean differences of about 1.15 standard deviations of the paired differences, 5.4 points for the final comparison with EEGNet (Section VI-L), so model differences of a few points cannot be resolved; the findings may also not carry over to other montages, paradigms or populations. That accuracy does not depend on model size is shown only for the architectures and sizes tested here, on 5,184 trials; for configurations A and D, the 95% interval rules out an advantage of more than about one point for the larger model.

Mixed protocols and baseline strength. ATCNet, EEG Conformer, CTNet and FBCSP were run only under protocol v1, with one seed and no test-time adaptation, whereas EEGNet and HCT-Net were also run under v2. The comparison of model sizes above 25,000 parameters therefore rests on the protocol we later found to be biased, and on single runs. Two of the large baselines did not reproduce their published within-subject accuracy under our shared recipe, so their cross-subject results may understate what these architectures can do. Comparisons that cross the two protocols indicate scale only; they are not rankings.

Shared training recipe. All deep models used the same optimiser, learning-rate schedule, epoch budget and early-stopping rule, and each baseline kept its default architectural hyperparameters. This is the fair comparison the study was designed to make, and it reflects calibration-free deployment, where no model can be tuned on the target user; it may, however, put at a disadvantage models whose published results rely on longer training or stronger regularisation. Under this recipe ATCNet and CTNet matched their published within-subject accuracy on decodable participants but not on weak ones.

Analysis window. All models received 3.5 s of signal (the interval 0.5–4.0 s following the cue), while some published models use 4–4.5 s. A planned experiment in which each model uses its own window had not finished when this paper was written.

Transductive adaptation. Alignment and batch-normalisation statistics were computed from all 576 unlabelled trials of the new user before any prediction was made. An online system receives trials one by one, and its gains at the start of a session would probably be smaller [29]. Because batch-normalisation adaptation was applied only under protocol v2, it also favours EEGNet and HCT-Net over the baselines evaluated under v1.

Order of experiments. Two choices were influenced by test results we had already seen. Global attention (C2) was tried partly because the component study, which reports test-fold accuracy, favoured it, and the test-time adaptation experiment was brought forward after an earlier test result looked encouraging. In both cases the keep-or-reject decision relied on validation only, and the adaptation settings were fixed beforehand; C2 is not part of the final model. The choice of what to try, however, was not made blind.

Logging gaps and post hoc analysis. Per-trial predictions were not saved during the size sweep, so the confusion analysis relies on the v1 models and no confusion matrix exists for the final configuration. For the same reason the cross-subject score could not be restricted to session E, and every familiar-versus-unseen gap compares accuracy on session E with accuracy on both sessions. The analysis of session shift was not part of the design and should be treated as exploratory, as should the confidence intervals, power analysis, nested selection and Riemannian baselines, which were added after the main study.

Non-determinism. Training on a GPU is not exactly reproducible. In our repeated runs with an unchanged seed most folds gave identical results and a few stopped at a different epoch; the resulting shifts were smaller than the seed noise reported in Table XIII.

IX. Conclusion and Future Work

We asked whether additional model capacity helps to decode the motor imagery of people a model has never seen, and studied the question under a protocol planned in advance, with its deviations reported, in which design choices were made on a held-out validation subject. On BCI Competition IV-2a we found no evidence that it does. Decoders with 3,188 to 697,412 parameters reached similar cross-subject accuracy under our first protocol, HCT-Net configurations from 3,924 to 24,836 parameters did so under the corrected one, and the proposed convolution–Transformer hybrid lost nothing by becoming smaller and simpler. What did help were two label-free operations that need nothing but unlabelled trials of the new user, Euclidean alignment and batch-normalisation re-estimation, together with a protocol that keeps early stopping apart from selection. With 3,924 parameters, the final HCT-Net reaches 53.6% four-class accuracy ($\kappa = 0.381$) on unseen users without calibration, on a par with EEGNet and ahead of the best Riemannian baseline by 3.1 points. A gap of about 18 points to within-subject accuracy remains at every model size. Most of the progress from the initial design to the final model came from the corrected protocol and from adaptation rather than from the architecture, and the limits set out in Section VIII—nine subjects, large baselines run only under the first protocol, and an untested encoder ablation—bound how far these conclusions reach.

Four directions follow from these results. The first is completing the comparison: running every baseline under protocol v2 with adaptation, testing configuration A without its encoder, giving each model its own analysis window, and adding the eight-subject, fixed-epoch protocol of earlier work [8] as a separate condition. The second is more subjects: pre-training on large multi-subject collections such as the 109-subject PhysioNet motor movement/imagery dataset [53], [54], and testing on BCI IV-2b and the High-Gamma dataset [9], would show whether accuracy stays flat in model size as the number of training people grows. The third is online label-free adaptation that updates alignment and normalisation statistics as each trial arrives [29]. The fourth is alignment beyond second-order statistics, targeting the session-level variability that predicts failure.

Data and code availability. The raw recordings with their SHA-256 checksums, the complete code, the per-fold result logs with their configurations, and, in the folder paper/, the scripts that reproduce every table and figure of this paper are available at github.com/sathvikparangi907-web/neural-decoder-eeg-motor-imagery. The dataset itself is distributed by BNCI Horizon 2020 under a CC BY-ND 4.0 licence.

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APPENDIX A
PER-SUBJECT RESULTS

Table XXIV and Table XXV give the per-subject accuracies behind the means reported in Section VI, so that every comparison can be checked subject by subject. Fig. 22 and Table XXIX give the class-level results of the six models compared under protocol v1.

TABLE XXIV
WITHIN-SUBJECT ACCURACY (%) PER PARTICIPANT (SESSION T → E)

Model	A01	A02	A03	A04	A05	A06	A07	A08	A09	Mean
EEG Conformer	89.9	63.1	90.9	69.2	67.4	57.4	90.3	87.2	80.3	77.3
HCT-Net D	83.7	56.9	89.8	66.4	63.5	55.9	85.1	81.2	76.5	73.2
EEGNet	74.3	49.3	86.0	52.8	56.6	41.1	80.2	76.6	76.5	65.9
FBCSP + LDA	76.4	46.0	86.7	59.8	50.7	41.5	75.3	74.1	71.4	64.7
CTNet	70.1	49.6	89.1	52.4	50.3	35.9	71.2	83.3	75.6	64.2
ATCNet	71.2	52.9	87.0	49.3	43.4	39.6	67.0	81.9	72.8	62.8
HCT-Net A	86.0	54.4	91.1	65.0	60.2	47.5	86.9	82.4	75.4	72.1
HCT-Net B	85.3	54.1	91.9	62.6	59.0	49.9	88.7	80.5	75.3	71.9
HCT-Net C	85.1	57.7	89.1	63.1	64.9	52.7	86.6	83.9	77.3	73.4

First six rows: one run each, fixed 500-epoch schedule. HCT-Net A–C: means over three seeds with the same recipe (size sweep).

TABLE XXV

CROSS-SUBJECT TEST ACCURACY (%) PER TEST SUBJECT UNDER PROTOCOL V2 (THREE-SEED MEANS)

Configuration	A01	A02	A03	A04	A05	A06	A07	A08	A09	Mean
HCT-Net A + BN adaptation	72.2	31.4	76.7	44.6	37.6	45.0	51.6	65.0	58.5	53.6
HCT-Net A, no adaptation	70.5	30.1	75.7	37.9	33.6	37.4	39.9	61.2	57.8	49.3
HCT-Net B + BN adaptation	72.7	31.1	78.0	43.8	38.2	45.5	49.1	63.8	59.6	53.5
HCT-Net C + BN adaptation	71.3	34.7	74.1	41.4	37.3	44.2	48.9	64.1	56.4	52.5
HCT-Net D + BN adaptation	72.5	34.0	76.8	42.7	36.2	46.9	46.5	63.1	55.6	52.7
HCT-Net C2 + BN adaptation	74.9	31.1	78.1	42.4	39.0	48.4	46.8	63.5	65.6	54.4
EEGNet + BN adaptation	74.4	30.7	77.3	46.1	36.6	48.8	42.5	65.3	67.0	54.3
EEGNet, no adaptation	70.2	31.2	77.6	42.9	34.0	47.2	37.9	65.7	63.1	52.2
TS-LR + re-centring (deterministic)	60.4	31.8	73.4	45.3	33.3	39.9	41.7	68.4	60.1	50.5
MDM + re-centring (deterministic)	59.6	22.9	59.4	43.8	31.2	36.6	47.7	65.3	57.8	47.1
MDM, no re-centring (deterministic)	34.9	23.6	54.2	26.2	25.0	22.7	35.2	25.4	25.0	30.2

Each fold's validation subject is the next subject (A09's is A01). The Riemannian baselines are trained on the same seven subjects per fold and use no validation subject.

Table XXVI gives the per-subject accuracies of the component study, the depth variants and the adversarial variant, Table XXVII those of the single-seed improvement round under protocol v1, and Table XXVIII the validation-subject accuracies on which the final selection was based.

TABLE XXVI

COMPONENT STUDY AND VARIANTS: CROSS-SUBJECT ACCURACY (%) PER TEST SUBJECT, PROTOCOL V1

Variant	A01	A02	A03	A04	A05	A06	A07	A08	A09	Mean
Full model (HCT-Net D)	72.7	28.3	70.1	34.7	34.4	33.0	34.9	62.0	52.1	46.9
V1 no attention	65.1	28.5	73.3	44.8	34.7	36.5	35.4	64.6	56.3	48.8
V2 global attention	70.1	28.0	78.6	38.0	33.3	42.0	33.9	64.9	59.5	49.8
V3 no alignment	57.5	26.0	58.9	37.5	28.1	27.4	29.5	58.2	53.5	41.8
V4 no augmentation	70.8	27.8	73.3	39.2	34.2	43.2	33.9	59.5	51.2	48.1
V5 convolution only	47.7	26.9	43.6	35.6	28.3	28.6	27.4	60.2	53.6	39.1
Adversarial (gradient reversal)	72.9	33.2	74.3	36.6	30.4	32.8	40.6	63.7	55.0	48.8
Depth 1 (L1)	66.5	27.4	78.1	34.7	34.5	31.9	32.8	56.4	51.6	46.0
Depth 6 (L6)	67.9	28.8	75.2	40.8	37.8	31.8	38.7	59.7	48.8	47.7

TABLE XXVII

SINGLE-SEED IMPROVEMENT ROUND: TEST ACCURACY (%) PER TEST SUBJECT, PROTOCOL V1

Configuration	A01	A02	A03	A04	A05	A06	A07	A08	A09	Mean
D (initial design)	71.2	30.7	72.6	35.1	32.8	30.6	39.8	60.1	50.9	47.1
C1 flatten head	66.3	33.7	74.0	36.6	34.5	37.2	38.9	63.7	51.9	48.5
C2 + global attention	67.9	31.6	75.9	36.6	31.6	39.1	33.2	60.8	64.9	49.1
C3 + BN adaptation	72.0	31.8	76.2	42.4	35.4	43.8	46.5	64.1	66.5	53.2
C5 three-seed ensemble	71.9	31.3	79.7	40.8	33.3	43.4	38.0	63.5	65.6	51.9

TABLE XXVIII

VALIDATION ACCURACY (%) PER FOLD, PROTOCOL V2 WITH BN ADAPTATION (THREE-SEED MEANS)

Fold (validation subject)	F1 (A02)	F2 (A03)	F3 (A04)	F4 (A05)	F5 (A06)	F6 (A07)	F7 (A08)	F8 (A09)	F9 (A01)	Mean
HCT-Net A	33.8	79.0	43.6	35.6	45.5	51.6	67.1	53.6	79.0	54.3
HCT-Net B	32.3	81.0	43.0	35.4	44.3	50.4	66.3	48.6	77.3	53.2
HCT-Net C	32.9	76.5	41.0	37.0	44.5	49.2	65.2	55.1	76.7	53.1
HCT-Net D	33.6	78.2	41.3	35.5	45.8	49.9	66.8	55.2	73.6	53.3
EEGNet	31.1	80.5	43.5	38.7	48.4	53.8	67.2	59.3	72.8	55.0

Fold k tests subject k and selects on subject $k + 1$. These are the numbers on which every protocol-v2 decision was taken.

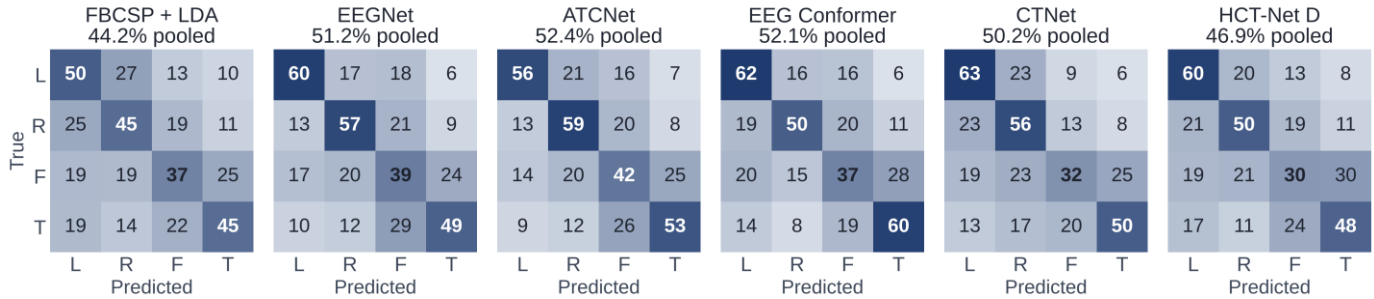


Fig. 22. Pooled cross-subject confusion matrices of the six models compared under protocol v1 (percentage of each true class; L: left hand, R: right hand, F: feet, T: tongue; 1,296 trials per class, one seed). Feet has the lowest recall for every model.

TABLE XXIX
PER-CLASS PRECISION, RECALL AND F1 (%) UNDER PROTOCOL V1, POOLED OVER THE NINE FOLDS

Model	Metric	Left	Right	Feet	Tongue	Mean
FBCSP + LDA	Precision	44.4	42.7	40.5	49.4	44.3
	Recall	50.0	45.0	37.3	44.5	44.2
	F1	47.1	43.8	38.8	46.9	44.1
EEGNet	Precision	59.7	53.9	36.8	55.6	51.5
	Recall	60.1	56.9	39.1	48.5	51.2
	F1	59.9	55.4	37.9	51.8	51.3
ATCNet	Precision	60.9	53.3	39.9	56.9	52.8
	Recall	56.3	59.2	41.5	52.7	52.4
	F1	58.5	56.1	40.7	54.7	52.5
EEG Conformer	Precision	54.1	56.2	40.0	57.3	51.9
	Recall	62.0	49.8	37.1	59.6	52.1
	F1	57.8	52.8	38.5	58.4	51.9
CTNet	Precision	53.5	46.9	43.5	56.0	50.0
	Recall	62.6	55.9	32.4	50.0	50.2
	F1	57.7	51.0	37.2	52.8	49.7
HCT-Net D	Precision	51.3	49.2	35.2	49.8	46.4
	Recall	59.6	49.9	30.3	47.8	46.9
	F1	55.1	49.5	32.6	48.8	46.5

Mean: unweighted mean over the four classes; for recall it equals the pooled accuracy, because every class has 1,296 test trials. Feet has the lowest F1 of the four classes for every model.

APPENDIX B
BASELINE REPRODUCTION DIAGNOSTICS

This appendix documents the diagnostic runs that came before the cross-subject experiments, since each of them changed how a baseline was run. These runs are recorded in the project's findings log rather than in the per-fold result files. Our first FBCSP implementation reached 61.6% within subject. Instead of guessing, we tested three possible causes (Table XXX): a shorter analysis window lowered accuracy, keeping the artefactual training trials changed almost nothing, and the size of the mutual-information feature budget turned out to be responsible. The final budget of 16 features was not tuned; it is the effective budget of the reference algorithm. The result for the window deserves a comment: the late part of the trial, when the mu rhythm has started to recover, still holds information for a classifier based on signal variance.

TABLE XXX
FBCSP DIAGNOSTICS (WITHIN SUBJECT, %)

Change tested	Accuracy	Verdict
First implementation	61.6	reference
Window 0.5–2.5 s instead of 0.5–4.0 s	59.0	refuted
All training trials kept	61.2	refuted
Feature budget $k = 4$	56.3	—
Feature budget $k = 8$	61.2	—
Feature budget $k = 16$	65.0	confirmed

Diagnostic runs during implementation; the final within-subject run with $k = 16$ gave 64.7% (Table VI).

ATCNet and CTNet initially scored about 18 points under their published within-subject results, so we tested three explanations (Table XXXI). None of them held, and the shortfall turned out to be limited to the weakest subjects. For EEGNet, the cause of its early shortfall was the 20% hold-out of within-subject training data (Table XXXII): using the final annealed weights rather than the best checkpoint changed little, whereas training on all 288 trials of session T improved every subject we tested.

TABLE XXXI
ATCNET AND CTNET DIAGNOSTICS (WITHIN SUBJECT)

Hypothesis	Test	Outcome
Implementation error	compared braindecode 1.8.1 with both papers	refuted: reference code, paper defaults
Undertrained	1,500 instead of 500 epochs	refuted: ATCNet 80.0 → 77.6, CTNet 80.8 → 80.4 (A01, A03, A08)
Input window too short	ATCNet at 1,125 samples	refuted: 43.9 → 41.1 on weak subjects

On A01, A03 and A08 both models reproduce their published means (80.0% and 80.8% against 81.1% and 82.5%); on A04, A05 and A06 ATCNet scores 49.3%, 43.4% and 39.6% (Table XXIV). The 1,125-sample comparison (43.9% against 41.1% on these three subjects) comes from a separate diagnostic run.

TABLE XXXII
EEGNET WITHIN SUBJECT: EFFECT OF THE 20% HOLDOUT (%)

Training data and checkpoint	A05	A06	A01	A08
80% of T, best checkpoint	26.7	35.6	70.5	65.2
80% of T, final annealed model	31.6	33.7	68.4	69.5
All of T, final annealed model	56.6	42.2	74.3	77.0

Diagnostic run on four subjects; the final within-subject protocol uses the last row's setting for all subjects.

APPENDIX C
TRAINING CONFIGURATION

TABLE XXXIII
TRAINING CONFIGURATION SHARED BY EVERY DEEP MODEL

Setting	Value
Input	22 × 875 (channels × samples), 0.5–4.0 s post-cue
Band-pass	4–38 Hz, 4th-order Butterworth, zero phase
Alignment	Euclidean, per session, all subjects
Normalisation	per channel, fit-set statistics
Augmentation	S&R, 8 segments, fit set doubled
Loss	cross-entropy, label smoothing 0.1
Optimiser	Adam, lr 10^{-3}, weight decay 10^{-4}
Schedule	cosine annealing over 500 epochs
Batch size	64
Early stopping	patience 50 on stop-set accuracy (v2)
Stop-set	one tenth of the training trials, stratified by subject and class
Seeds	0, 1, 2 (v2)
Adaptation	BN statistics re-estimated on target (bn)
Software	PyTorch 2.14, braindecode 1.8.1, Python 3.14
Hardware	NVIDIA RTX 4050 laptop GPU, 6 GB

APPENDIX D
REPRODUCIBILITY CHECKLIST

Table XXXIV maps each item of a standard reproducibility checklist to the place in this paper, or in the released repository, where it is specified.

TABLE XXXIV
REPRODUCIBILITY CHECKLIST

Item	Where it is specified
Dataset and version	BNCI Horizon 001-2014 .mat files; 18 SHA-256 checksums in the repository
Inclusion and exclusion	all 9 subjects, both sessions; flagged trials removed from training subjects only
Preprocessing	Section III-B, Section IV-B; one code path for every model
Folds and roles	Section V-B, Fig. 8; fold k tests subject k, validates on $k + 1$
Model definitions	Table IV, Table V; braindecode 1.8.1 defaults for baselines
Training recipe	Section IV-E, Table XXXIII
Seeds and repetitions	three seeds per configuration under protocol v2
Decision rules	Section V-D; fixed before each experiment
Statistics	Section V-E; two-sided Wilcoxon signed-rank test (SciPy), Holm correction; 95% intervals, detectable effects and nested selection in Section VI-L and Section VI-J
Riemannian baselines	script riemann_loso.py; deterministic, same folds and preprocessing
Per-fold results	one CSV row per fold and seed, with configuration file
Compute	one NVIDIA RTX 4050 (6 GB); about 36 h of logged training
Negative results	reported in Section VI and Table XXIII

APPENDIX E
ABBREVIATIONS

Table XXXV lists the abbreviations used in the text, tables and figures.

TABLE XXXV
ABBREVIATIONS

Abbreviation	Meaning
BCI	brain–computer interface
BN	batch normalisation
CI	confidence interval
CSP / FBCSP	common spatial patterns / filter-bank CSP
EA	Euclidean alignment
EEG / EOG	electroencephalography / electro-oculography
ERD	event-related desynchronisation
ELU / GELU	exponential / Gaussian error linear unit
FFN	feed-forward network
GRL	gradient reversal layer
HCT-Net	hybrid convolution–Transformer network (this work)
IQR	interquartile range
ITR	information transfer rate
LDA	linear discriminant analysis
LOSO	leave-one-subject-out
MDE	minimum detectable effect
MDM	minimum distance to (Riemannian) mean
MI	motor imagery (also: mutual information, in FBCSP)
RC	Riemannian re-centring
S&R	segmentation and reconstruction (augmentation)
TENT	test-time entropy minimisation
TS-LR	tangent-space logistic regression
TTA	test-time adaptation

APPENDIX F
NOTATION

Table XXXVI lists the symbols used in the equations and their meaning.

TABLE XXXVI
SYMBOLS USED IN THIS PAPER

Symbol	Meaning
$X \in \mathbb{R}^{C\times T}$	one trial: $C = 22$ channels, $T = 875$ samples
y, k	class label, class index (1–4)
R_s	mean spatial covariance of session s
Z_i	trial after Euclidean alignment
μ_c, σ_c	channel mean and standard deviation of the fit set
F_1, D, F_2	temporal filters, spatial depth multiplier, separable filters
d, d_h	encoder model dimension (= F_2) and head dimension
W, L	number of attention windows (5) and window length (11 steps)
P	learned positional code, shared across windows
$H^{(w)}$, $E^{(w)}$	input and output of the encoder for window w
f, p	pooled feature vector, predicted class probabilities
ε	label-smoothing factor (0.1)
μ, σ^2, γ, β	batch-normalisation statistics and affine parameters
Σ_k	average covariance of class k (CSP)
κ	Cohen's kappa
B, P_c, M	information transfer rate (bits per selection), probability of a correct selection, number of classes